

HW Assignment 11-1 (11.1, 11.3, 11.4, 11.6)**11.1 (a)****Homogeneity**

Since the space is homogeneous, the origin of a frame is arbitrary. The transformation should not depend on the location of the origin, even with a change of the origin. Consider a linear transformation between K and K'

$$x' = c_{11}x + c_{12}t \quad \text{and} \quad t' = c_{21}x + c_{22}t \quad (1)$$

where c_{ij} are independent of (x, t) . For a change of the origins $(x, t) \rightarrow (x - x_0, t - t_0)$ and $(x', t') \rightarrow (x' - x'_0, t' - t'_0)$, the transformation becomes

$$x' - x'_0 = c_{11}(x - x_0) + c_{12}(t - t_0) \quad \text{and} \quad t' - t'_0 = c_{21}(x - x_0) + c_{22}(t - t_0) \quad (2)$$

which is the same transformation in Eq. (1) with a consistent transformation between the origins as

$$x'_0 = c_{11}x_0 + c_{12}t_0 \quad \text{and} \quad t'_0 = c_{21}x_0 + c_{22}t_0$$

Therefore, a linear transformation is independent of the choose of the origins of the frames. It is no longer true for nonlinear transformations between K and K' . A nonlinear transformation of

$$x' = f_1(x, t) \quad \text{and} \quad t' = f_2(x, t) \quad (3)$$

becomes

$$x' - x'_0 = f_1(x - x_0, t - t_0) \quad \text{and} \quad t' - t'_0 = f_2(x - x_0, t - t_0) \quad (4)$$

after a change of the origins. The transformation in Eq. (4) always depends on (x_0, t_0) or the transformation of between (x'_0, t'_0) and (x_0, t_0) depends on (x, t) . Therefore, the space is not homogeneous anymore.

Isotropy

For a linear transformation in Eq. (1), c_{ij} are functions of v , the velocity of K' moving in K along the x direction. Because the space is isotropic, a change of the direction of v should not change the direction of t and a change of the direction of v corresponds to a change of the direction of x and x' , therefore, c_{11} and c_{22} are even functions of v and c_{12} and c_{21} are odd functions of v , *i.e.*

$$x' = f_1(v^2)x - vf_2(v^2)t \quad \text{and} \quad t' = g(v^2)t - vh(v^2)x \quad (5)$$

Location of the Origin of K' Frame in K Frame

In K frame, the origin of K' ($x' = 0$) is moving with velocity v and located at $x = vt$. From the transformation in Eq. (5), $f_1(v^2)x - vf_2(v^2)t = 0$ for $x' = 0$, which leads to

$$f_1(v^2) = f_2(v^2) = f(v^2)$$

Moreover, because of the symmetry between K and K' in term of the direction of v , the inverse transformation of Eq. (5) should be

$$x = f(v^2)x' + vf(v^2)t' \quad \text{and} \quad t = g(v^2)t' + vh(v^2)x' \quad (6)$$

11.1 (b)

The linear transformation in Eqs. (5) and (6) can be written in matrix form as

$$\begin{pmatrix} x' \\ t' \end{pmatrix} = \begin{pmatrix} f & -vf \\ -vh & g \end{pmatrix} \begin{pmatrix} x \\ t \end{pmatrix} \quad \text{and} \quad \begin{pmatrix} x \\ t \end{pmatrix} = \begin{pmatrix} f & vf \\ vh & g \end{pmatrix} \begin{pmatrix} x' \\ t' \end{pmatrix}$$

Therefore,

$$\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} f & -vf \\ -vh & g \end{pmatrix} \begin{pmatrix} f & vf \\ vh & g \end{pmatrix} = \begin{pmatrix} f(f - v^2h) & vf(f - g) \\ vh(g - f) & g^2 - v^2fh \end{pmatrix}$$

which leads

$$f = g \quad \text{and} \quad f(f - v^2h) = 1$$

11.1 (c)

The transformation in Eq. (6) is then

$$x = fx' + vft' \quad \text{and} \quad t = ft' + vhx'$$

or

$$dx = fdx' + vfdt' \quad \text{and} \quad dt = fdt' + vhd x'$$

Then

$$u = \frac{dx}{dt} = \frac{fdx' + vfdt'}{fdt' + vhd x'} = \frac{u' + v}{1 + vu'(h/f)}$$

Postulate II. The maximal speed of any object is c

If $u' = c$ and $v = c$, Postulate II requires $u = c$, *i.e.*

$$c = \frac{c + c}{1 + c^2(h/f)} \quad \implies \quad \frac{h}{f} = \frac{1}{c^2}$$

$$f(f - v^2h) = 1 \quad \implies \quad f = 1/\sqrt{1 - v^2/c^2} = \gamma$$

11.3

The Lorentz transformation for $(x_0, x_1) \longrightarrow (x'_0, x'_1)$ when $\vec{\beta} \parallel \vec{e}_x$ can be written as

$$\begin{pmatrix} x'_0 \\ x'_1 \end{pmatrix} = \begin{pmatrix} \gamma & -\gamma\beta \\ -\gamma\beta & \gamma \end{pmatrix} \begin{pmatrix} x_0 \\ x_1 \end{pmatrix}$$

Two successive transformations for $(x''_0, x''_1) \longrightarrow (x'_0, x'_1) \longrightarrow (x_0, x_1)$ is then

$$\begin{pmatrix} x''_0 \\ x''_1 \end{pmatrix} = \begin{pmatrix} \gamma_2 & -\gamma_2\beta_2 \\ -\gamma_2\beta_2 & \gamma_2 \end{pmatrix} \begin{pmatrix} \gamma_1 & -\gamma_1\beta_1 \\ -\gamma_1\beta_1 & \gamma_1 \end{pmatrix} \begin{pmatrix} x_0 \\ x_1 \end{pmatrix} = A \begin{pmatrix} x_0 \\ x_1 \end{pmatrix}$$

where

$$A = \begin{pmatrix} \gamma_1\gamma_2(1 + \beta_1\beta_2) & -\gamma_1\gamma_2(\beta_1 + \beta_2) \\ -\gamma_1\gamma_2(\beta_1 + \beta_2) & \gamma_1\gamma_2(1 + \beta_1\beta_2) \end{pmatrix} = \begin{pmatrix} \gamma & -\gamma\beta \\ -\gamma\beta & \gamma \end{pmatrix}$$

where

$$\gamma = \gamma_1\gamma_2(1 + \beta_1\beta_2) \quad \text{and} \quad \beta = \frac{\beta_1 + \beta_2}{1 + \beta_1\beta_2}$$

To verify the consistency of the definition of β and γ ,

$$\begin{aligned} \gamma^2 &= \frac{1}{1 - \beta^2} = \frac{(1 + \beta_1\beta_2)^2}{(1 + \beta_1\beta_2)^2 - (\beta_1 + \beta_2)^2} \\ &= \frac{(1 + \beta_1\beta_2)^2}{(1 - \beta_1^2)(1 - \beta_2^2)} = \gamma_1^2\gamma_2^2(1 + \beta_1\beta_2)^2 \end{aligned}$$

11.4

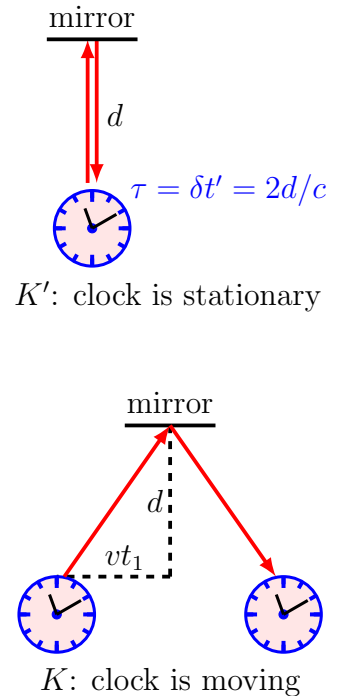
In the clock rest frame K' , the light is emitted from flashtube F , bounced at mirror M , and received at photocell P . The time for this light path observed in K' is $\tau = 2d/c$.

(a) Consider that the clock is moving horizontally (perpendicular to PF to M) with a velocity v in Lab frame K . In K , after the light is emitted from the tube, the tube F and the mirror M is moving forward horizontally to F' and M' , respectively, when the light reaches the mirror. The light path observed in the lab frame is along the hypotenuse of the right triangle $FM'F'$. Let t_1 be the time for the light travel from the tube to the mirror. Then

$$ct_1 = \sqrt{d^2 + (vt_1)^2} \quad \text{or} \quad t_1 = \frac{d/c}{\sqrt{1 - v^2/c^2}}$$

Similarly, the time for the light travel from the mirror to the photocell observed in the lab frame is also t_1 . The total time for one clock tick observed in the Lab frame is thus

$$t = t_1 + t_1 = \frac{\tau}{\sqrt{1 - v^2/c^2}} = \gamma\tau$$



(b) Consider that the clock is moving upward (parallel to PF to M) with velocity v in the Lab frame K . In K frame, after the light is emitted from the tube, the mirror is moving away from the light. If the time for the light reaching the mirror is t_1 , then

$$ct_1 = d + vt_1 \quad \text{or} \quad t_1 = \frac{d}{c - v}$$

If the time for the light travel from M to P , then

$$ct_2 = d - vt_2 \quad \text{or} \quad t_2 = \frac{d}{c + v}$$

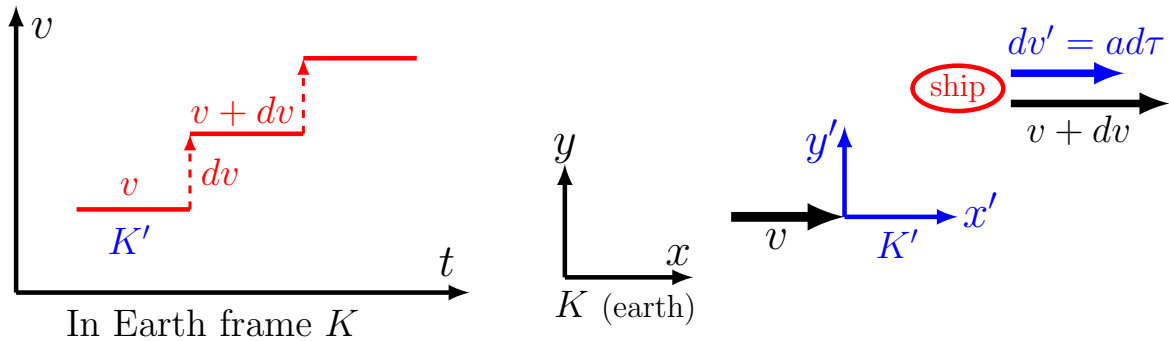
The total time of one clock tick observed in K is then

$$t = t_1 + t_2 = \frac{2d/c}{1 - v^2/c^2} = \frac{\tau}{1 - v^2/c^2} = \gamma^2 \tau$$

Therefore, the clock observed in the rest (proper) frame is ticking even more slowly by an additional γ factor. This additional γ factor is due to the length contraction of an moving object along the direction of $\vec{\beta}$ observed in the Lab frame.

11.6 Twins Paradox — How to Consider Acceleration of Spaceship

A spaceship is moving away from the Earth with an acceleration. Since the spaceship is not an inertial frame in general, the proper time (time on a stationary clock) observed on the spaceship is not symmetric with the proper time on the Earth. To use the formula of the proper time, $dt = \gamma d\tau$, to calculate to the difference of the time elapsed on the Earth and on the spaceship, we need first to calculate the time dependence of the spaceship velocity, $\beta = \beta(t)$, observed on the earth when the spaceship is accelerating away from the earth. In the case of a slow acceleration, one can use the adiabatic approximation in which the acceleration can be treated as discrete steps of velocity increase. Between two steps of the velocity, the system are in inertial frames and the velocity change from one step to another can be calculated using the Lorentz transformation for the velocity.



In Earth frame K ,	the spaceship velocity is $u_x = v + dv$,	time observed is t
In Ship frame K' ,	the spaceship velocity is $u'_x = dv'$,	time observed is τ

Consider a spaceship at a given time t is moving with velocity v and accelerated to $v + dv$ as observed from the Earth, where $dv > 0$ for the case of acceleration and $dv < 0$ for the case of deceleration. Let K be the Earth reference frame and K' be the reference frame moving with the spaceship at velocity v where v changes with each step of the spaceship velocity. After one step of the velocity change, $v \rightarrow v + dv$ in the Earth frame K and $0 \rightarrow dv'$ in the ship frame K' , where K' frame is moving with v with respect to K frame along the x direction. The Lorentz transformation between the velocity observed in K frame and observed in K' frame is

$$u_x = \frac{v + u'_x}{1 + vu'_x/c^2} \quad \Rightarrow \quad v + dv = \frac{v + dv'}{1 + vdv'/c^2}$$

Consider a constant acceleration when observed in the spaceship as the acceleration is controlled (and measured) by the driver of the spaceship. When observed from the Earth, the acceleration is not constant and the time dependence of the acceleration needs to be calculated. Let a be the constant acceleration observed in K' frame. For an infinitesimal increase of the velocity in K' frame, $dv' = ad\tau \ll v$ and

$$v + dv = \frac{v + ad\tau}{1 + vad\tau/c^2} \simeq (v + ad\tau) \left(1 - \frac{va}{c^2}d\tau\right) \simeq v + \left(1 - \frac{v^2}{c^2}\right)ad\tau$$

Note that this equation can be use for both cases of acceleration with $a > 0$ and deceleration with $a < 0$. With $dt = d\tau/\gamma$, it yields

$$dv = (1 - \beta^2)ad\tau = \frac{(1 - \beta^2)a}{\gamma}dt \quad \Rightarrow \quad \frac{\gamma d\beta}{(1 - \beta^2)} = \frac{adt}{c}$$

where $\beta = v/c$ and $d\tau = dt/\gamma$, and

$$\int_0^t \frac{at}{c} = \int_{\beta_0}^{\beta} \frac{d\beta}{(1 - \beta^2)^{3/2}}$$

where $\beta_0 = v_0/c$ and v_0 is the initial velocity of the spaceship. To simplify the calculation, we choose $v_0 = 0$. To calculate the integral let $\beta = \cos \theta$,

$$\frac{at}{c} = - \int \frac{d \cos \theta}{\sin^3 \theta} = \int \frac{d\theta}{\sin^2 \theta} = \frac{\cos \theta}{\sin \theta} = \frac{\beta}{\sqrt{1 - \beta^2}}$$

The velocity as a function of time is then

$$\beta(t) = \sqrt{\frac{(at/c)^2}{1 + (at/c)^2}} \quad \text{and} \quad \gamma(t) = \sqrt{1 + (at/c)^2}$$

where the velocity of the spaceship can never exceed c no matter how large the acceleration or how long the time τ is. The relationship between the time observed on the Earth t and on the spaceship τ is

$$d\tau = \frac{dt}{\gamma} = \frac{dt}{\sqrt{1 + (at/c)^2}}$$

and

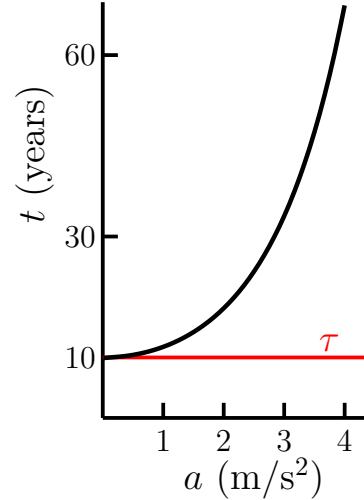
$$\tau = \int \frac{dt}{\sqrt{1 + (at/c)^2}} = \frac{c}{a} \int \frac{dx}{\sqrt{1 + x^2}} = \frac{c}{a} \sinh^{-1} \frac{at}{c}$$

or

$$\frac{at}{c} = \sinh \frac{a\tau}{c}$$

Since (10 year) $\simeq 3 \times 10^8 s$, for $\tau = 10$ year, $\tau/c \simeq 1s^2/m$ and

$$t = \left(\frac{1m/s^2}{a} \sinh \frac{a}{1m/s^2} \right) \times (10 \text{ years})$$



For example, for $a = 4 m/s^2$, $t = 68$ years while $\tau = 10$ years. Note that this formula can be use for the both cases of acceleration with $a > 0$ and deceleration with $a < 0$. For a spaceship that travels from the Earth to somewhere far away in the universe and then comes back to the Earth, it's path consists of two segments of acceleration and two segments of deceleration. The four segments of acceleration or deceleration have an equal duration of time τ when observed on the spaceship. The total time observed on the earth is then

$$t = \frac{4c}{a} \sinh \frac{a\tau}{c}$$

For a spaceship 10 years (5 years acceleration and 5 years deceleration) to a point, the distance traveled by the spaceship observed on the earth is

$$d = \int_0^t v(t)dt = c \int_0^t \frac{at/c}{\sqrt{1 + (at/c)^2}} dt = \frac{c^2}{2a} \int_0^t \frac{d(at/c)^2}{\sqrt{1 + (at/c)^2}} = \frac{c^2}{2a} \left[\sqrt{1 + (at/c)^2} - 1 \right]$$

where

$$t = \frac{c}{a} \sinh \frac{a\tau}{c}$$

and $\tau/c = 5 \text{ years}/c = 0.5 s^2/m$.

HW Assignment 11-2 (Extra 1)

Extra HW 1. Prove Eq. (11.98) on page 547 of the textbook.

For Lorentz boost with arbitrary $\vec{\beta}$, $\mathbf{L} = -\vec{\zeta} \cdot \vec{\mathbf{K}}$. Let $\vec{\zeta} = \zeta \vec{\epsilon}$, where $|\vec{\epsilon}| = 1$. Then

$$\mathbf{A}_{boost}(\vec{\beta}) = e^{-\zeta \vec{\epsilon} \cdot \vec{\mathbf{K}}} = \sum_{n=0}^{\infty} \frac{(-\zeta)^n}{n!} (\vec{\epsilon} \cdot \vec{\mathbf{K}})^n$$

From HW 11.10, we have

$$\vec{\epsilon} \cdot \vec{\mathbf{K}} = \begin{pmatrix} 0 & \epsilon_1 & \epsilon_2 & \epsilon_3 \\ \epsilon_1 & 0 & 0 & 0 \\ \epsilon_2 & 0 & 0 & 0 \\ \epsilon_3 & 0 & 0 & 0 \end{pmatrix} = \begin{pmatrix} \mathbf{0}_{1 \times 1} & \boldsymbol{\epsilon}_{1 \times 3} \\ \boldsymbol{\epsilon}_{3 \times 1}^T & \mathbf{0}_{3 \times 3} \end{pmatrix}$$

where $\mathbf{0}_{l \times l}$ is a $l \times l$ zero matrix and $\boldsymbol{\epsilon}_{1 \times 3} = (\epsilon_1, \epsilon_2, \epsilon_3)$, and

$$(\vec{\epsilon} \cdot \vec{\mathbf{K}})^2 = \begin{pmatrix} \mathbf{I}_{1 \times 1} & \mathbf{0}_{1 \times 3} \\ \mathbf{0}_{3 \times 1} & \mathbf{C} \end{pmatrix}$$

where

$$\mathbf{C} = \boldsymbol{\epsilon}^T \boldsymbol{\epsilon} = \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \epsilon_3 \end{pmatrix} \begin{pmatrix} \epsilon_1 & \epsilon_2 & \epsilon_3 \end{pmatrix} = (\epsilon_i \epsilon_j)_{3 \times 3}$$

is a 3×3 matrix with $C_{ij} = \epsilon_i \epsilon_j$.

Since

$$(\vec{\epsilon} \cdot \vec{\mathbf{K}})^2 (\vec{\epsilon} \cdot \vec{\mathbf{K}})^2 = \begin{pmatrix} \mathbf{I}_{1 \times 1} & \mathbf{0}_{1 \times 3} \\ \mathbf{0}_{3 \times 1} & \mathbf{C} \end{pmatrix} \begin{pmatrix} \mathbf{I}_{1 \times 1} & \mathbf{0}_{1 \times 3} \\ \mathbf{0}_{3 \times 1} & \mathbf{C} \end{pmatrix} = \begin{pmatrix} \mathbf{I}_{1 \times 1} & \mathbf{0}_{1 \times 3} \\ \mathbf{0}_{3 \times 1} & \mathbf{C}^2 \end{pmatrix}$$

and

$$\mathbf{C}^2 = \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \epsilon_3 \end{pmatrix} \begin{pmatrix} \epsilon_1 & \epsilon_2 & \epsilon_3 \end{pmatrix} \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \epsilon_3 \end{pmatrix} \begin{pmatrix} \epsilon_1 & \epsilon_2 & \epsilon_3 \end{pmatrix} = \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \epsilon_3 \end{pmatrix} \begin{pmatrix} \epsilon_1 & \epsilon_2 & \epsilon_3 \end{pmatrix} = \mathbf{C}$$

where

$$\begin{pmatrix} \epsilon_1 & \epsilon_2 & \epsilon_3 \end{pmatrix} \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \epsilon_3 \end{pmatrix} = |\vec{\epsilon}|^2 = 1$$

$$(\vec{\epsilon} \cdot \vec{\mathbf{K}})^{2m} = \begin{pmatrix} \mathbf{I}_{1 \times 1} & \mathbf{0}_{1 \times 3} \\ \mathbf{0}_{3 \times 1} & \mathbf{C} \end{pmatrix}$$

From HW 11.10,

$$(\vec{\epsilon} \cdot \vec{\mathbf{K}})^2 (\vec{\epsilon} \cdot \vec{\mathbf{K}}) = (\vec{\epsilon} \cdot \vec{\mathbf{K}}) \implies (\vec{\epsilon} \cdot \vec{\mathbf{K}})^{2m+1} = (\vec{\epsilon} \cdot \vec{\mathbf{K}})$$

$$\begin{aligned}
\mathbf{A}_{boost}(\vec{\beta}) &= \sum_{n=0}^{\infty} \left[\frac{(-\zeta)^{2n}}{(2n)!} (\vec{\epsilon} \cdot \vec{\mathbf{K}})^{2n} + \frac{(-\zeta)^{(2n+1)}}{(2n+1)!} (\vec{\epsilon} \cdot \vec{\mathbf{K}})^{2n+1} \right] \\
&= \mathbf{I} + \sum_{n=1}^{\infty} \frac{\zeta^{2n}}{(2n)!} \begin{pmatrix} \mathbf{I}_{1 \times 1} & \mathbf{0}_{1 \times 3} \\ \mathbf{0}_{3 \times 1} & \mathbf{C} \end{pmatrix} - \sum_{n=0}^{\infty} \frac{\zeta^{2n+1}}{(2n+1)!} \begin{pmatrix} \mathbf{0}_{1 \times 1} & \boldsymbol{\epsilon}_{1 \times 3} \\ \boldsymbol{\epsilon}_{3 \times 1}^T & \mathbf{0}_{3 \times 3} \end{pmatrix} \\
&= \begin{pmatrix} \mathbf{0}_{1 \times 1} & \mathbf{0}_{1 \times 3} \\ \mathbf{0}_{3 \times 1} & \mathbf{I} - \mathbf{C} \end{pmatrix} + \cosh \zeta \begin{pmatrix} \mathbf{I}_{1 \times 1} & \mathbf{0}_{1 \times 3} \\ \mathbf{0}_{3 \times 1} & \mathbf{C} \end{pmatrix} - \sinh \zeta \begin{pmatrix} \mathbf{0}_{1 \times 1} & \boldsymbol{\epsilon}_{1 \times 3} \\ \boldsymbol{\epsilon}_{3 \times 1}^T & \mathbf{0}_{3 \times 3} \end{pmatrix} \\
&= \begin{pmatrix} \cosh \zeta & -(\sinh \zeta) \boldsymbol{\epsilon}_{1 \times 3} \\ -(\sinh \zeta) \boldsymbol{\epsilon}_{3 \times 1}^T & \mathbf{I} + (\cosh \zeta - 1) \mathbf{C} \end{pmatrix}
\end{aligned}$$

Let

$$\gamma = \cosh \zeta, \quad \gamma\beta = \sinh \zeta, \quad \zeta = \tanh^{-1} \beta, \quad \text{with} \quad \cosh^2 \zeta - \sinh^2 \zeta = 1,$$

and

$$\vec{\epsilon} = \vec{\beta}/\beta$$

Then

$$\mathbf{A}_{boost}(\vec{\beta}) = \begin{pmatrix} \gamma & -\gamma\beta_1 & -\gamma\beta_2 & -\gamma\beta_3 \\ -\gamma\beta_1 & & & \\ -\gamma\beta_2 & \mathbf{I} + (\gamma - 1)\mathbf{C} & & \\ -\gamma\beta_3 & & & \end{pmatrix}$$

where

$$C_{ij} = \beta_i \beta_j / \beta^2$$

HW Assignment 11-3 (11.22, 11.23)

11.22

Collisions of two photons can create electron-positrons pairs. When a high energy photon travels through the cosmic microwave background, it could be scattered in this process so that it cannot freely propagate across galactic distances. The minimal photon energy for producing a pair of electron-positron sets an upper limit to the energy of photons that can propagate through the universe.

The conservation of 4-vector momentum of this photon-photon collision process is

$$p_{before}^{\alpha} = p_{in}^{\alpha} + p_{bg}^{\alpha} = p_{e^{-}}^{\alpha} + p_{e^{+}}^{\alpha} = p_{after}^{\alpha}$$

where p_{in}^α and p_{bg}^α are the momentum of the incident photon and microwave-background photon, respectively, and $p_{e^\pm}^\alpha$ are the momentum of electron (-) and positron (+), respectively. Since photon is massless, its energy and momentum are

$$E = h\nu \quad \text{and} \quad p = \frac{h}{\lambda} \quad \implies \quad E = cp$$

and its 4-vector momentum of photon is

$$p^\alpha = (p^0, \vec{p}) = (p, \vec{p})$$

where $p = |\vec{p}|$ and $p^0 = E/c = p$. For electron/positron,

$$p^\alpha = (E/c, \vec{p}) \quad \text{and} \quad E = \sqrt{c^2 p^2 + m_e^2 c^4}$$

where m_e is the electron rest mass. For the minimal energy of the incident photon for creating an electron-positron pair, we are looking for a pair of electron-positron with the minimal energy. In the CM frame, this minimal energy corresponds to zero kinetic energy for both electron and positron, *i.e.*

$$E = m_e c^2 \quad \text{and} \quad p_{e^\pm}^\alpha = (E/c, 0, 0, 0)$$

In the CM frame, therefore,

$$p_{after}^\alpha = p_{e^-}^\alpha + p_{e^+}^\alpha = (2m_e c, 0, 0, 0)$$

To obtain the energy/momentum in the lab frame, we use the invariance of the scalar products of 4-vectors in Lorentz transformation. Since

$$\mathbf{p}_{before} \cdot \mathbf{p}_{before} = \mathbf{p}_{after} \cdot \mathbf{p}_{after} = (2m_e c)^2,$$

in the lab frame,

$$(\mathbf{p}_{in} + \mathbf{p}_{bg}) \cdot (\mathbf{p}_{in} + \mathbf{p}_{bg}) = (2m_e c)^2$$

To further evaluate this, we need to choose a specific coordinate. Consider that initially the incident and microwave-background photon are moving in the positive and negative x direction, respectively, in the lab frame, *i.e.*

$$\mathbf{p}_{in} = (p_1, p_1, 0, 0) \quad \text{and} \quad \mathbf{p}_{bg} = (p_2, -p_2, 0, 0)$$

where $p_1 = E_{in}/c$, $p_2 = E_{bg}/c$, E_{in} and E_{bg} are the energy of the incident and microwave-background photon, respectively. Then,

$$(p_1 + p_2)^2 - (p_1 - p_2)^2 = (2m_e c)^2 \implies 4p_1 p_2 = (2m_e c)^2 \implies E_{in} E_{bg} = m_e^2 c^4$$

Therefore, the minimal energy of the incident photon for the electron-positron production in the lab frame with a given microwave-background photon satisfy

$$E_{in} = \frac{m_e^2 c^4}{E_{bg}}$$

where $m_e c^2 = 0.511 \text{ MeV}$.

(a) For the current universe, the microwave background has a temperature of $T = 3 \text{ K}$ that corresponds to $E_{bg} = kT = 2.5 \times 10^{-4} \text{ eV}$,

$$E_{in} = \frac{m_e^2 c^4}{E_{bg}} = \frac{(0.511 \times 10^6)^2}{2.5 \times 10^{-4}} \text{ eV} = 1.04 \times 10^9 \text{ MeV}$$

The minimal wave length of a photon that can travel through the universe is then

$$E_{in} = h\nu_{max} = \frac{ch}{\lambda_{min}} \quad \Rightarrow \quad \lambda_{min} = \frac{ch}{E_{in}} = \frac{3 \times 10^8 \times 6.6 \times 10^{-34}}{10^{15} \times 1.6 \times 10^{-19}} \sim 10^{-21} \text{ m}$$

(b) If $E_{bg} = 500 \text{ eV}$ with $T \sim 10^7 \text{ K}$.

$$E_{in} = \frac{m_e^2 c^4}{E_{bg}} = \frac{(0.511 \times 10^6)^2}{500} \text{ eV} = 522 \text{ MeV} \quad \Rightarrow \quad \lambda_{min} \sim 10^{-15}$$

11.23 In a collision process a particle of mass m_2 , at rest in the Lab frame, is struck by a particle of mass m_1 , momentum $\vec{p}_{LAB}$ and total energy E_{LAB} . In the collision the two initial particles are transformed into two others of mass m_3 and m_4 .

The conservation of 4-vector momentum for the collision in the Lab frame is

$$p_1^\alpha + p_2^\alpha = p_3^\alpha + p_4^\alpha$$

and in the CM frame is

$$p_1'^\alpha + p_2'^\alpha = p_3'^\alpha + p_4'^\alpha$$

where p_1^α (and $p_1'^\alpha$) and p_2^α (and $p_2'^\alpha$) are the 4-momentum of particle 1 and 2 before the collision and p_3^α (and $p_3'^\alpha$) and p_4^α (and $p_4'^\alpha$) are the 4-momentum of particle 3 and 4 after the collision. The 4-vector scalar product of the total momentum before and after the collision is invariant under the Lorentz transformation, *i.e.*

$$\begin{aligned} \text{in the Lab frame:} \quad (p_1 + p_2)^\alpha (p_1 + p_2)_\alpha &= (p_3 + p_4)^\alpha (p_3 + p_4)_\alpha \\ &\parallel \\ \text{in the CM frame:} \quad (p_1' + p_2')^\alpha (p_1' + p_2')_\alpha &= (p_3' + p_4')^\alpha (p_3' + p_4')_\alpha \end{aligned}$$

In the lab frame,

$$\begin{cases} p_1^\alpha = (E_1/c, \vec{p}_1) = (E_{LAB}/c, \vec{p}_{LAB}) \\ p_2^\alpha = (E_2/c, \vec{p}_2) = (m_2 c, 0, 0, 0) \end{cases}$$

where $E_{LAB} = \gamma_{LAB} m_1 c^2$, $\gamma_{LAB} = \sqrt{1 - v_{LAB}^2/c^2}$, and

$$\begin{aligned} (p_1 + p_2)^\alpha (p_1 + p_2)_\alpha &= (\gamma_{LAB} m_1 c + m_2 c)^2 - p_{LAB}^2 \\ &= m_1^2 \gamma_{LAB}^2 c^2 + 2\gamma_{LAB} m_1 m_2 c^2 + m_2^2 c^2 - m_1^2 \gamma_{LAB}^2 v_{LAB}^2 \\ &= 2m_2 E_{LAB} + (m_1^2 + m_2^2) c^2 \end{aligned}$$

In the CM frame,

$$\begin{cases} p_1'^\alpha &= (E_1'/c, \vec{p}') \\ p_2'^\alpha &= (E_2'/c, -\vec{p}') \end{cases} \quad \text{and} \quad \begin{cases} p_3'^\alpha &= (E_3'/c, \vec{q}) \\ p_4'^\alpha &= (E_4'/c, -\vec{q}) \end{cases}$$

and

$$(p_1' + p_2')^\alpha (p_1' + p_2')_\alpha = \frac{1}{c^2} (E_1' + E_2')^2 = \frac{1}{c^2} W^2$$

where $W = E_1' + E_2'$ is the total energy in the CM frame.

23.(a)

$$\begin{aligned} W^2 &= c^2 (p_1' + p_2')^\alpha (p_1' + p_2')_\alpha = c^2 (p_1 + p_2)^\alpha (p_1 + p_2)_\alpha \\ &= 2m_2 c^2 E_{LAB} + (m_1^2 + m_2^2) c^4 \end{aligned}$$

Since

$$\begin{cases} (p_1' + p_2')^\alpha (p_1' + p_2')_\alpha &= \frac{1}{c^2} (E_1' + E_2')^2 = \frac{1}{c^2} (E_1'^2 + E_2'^2 + 2E_1' E_2') \\ (p_1 + p_2)^\alpha (p_1 + p_2)_\alpha &= 2m_2 E_{LAB} + (m_1^2 + m_2^2) c^2 \end{cases}$$

where $E_1'^2 = \sqrt{c^2 p'^2 + m_1^2 c^4}$ and $E_2'^2 = \sqrt{c^2 p'^2 + m_2^2 c^4}$

$$\begin{aligned} 2m_2 E_{LAB} + (m_1^2 + m_2^2) c^2 &= \frac{1}{c^2} (E_1'^2 + E_2'^2 + 2E_1' E_2') \\ &= m_1^2 c^2 + p'^2 + m_2^2 c^2 + p'^2 + 2\sqrt{m_1^2 c^2 + p'^2} \sqrt{m_2^2 c^2 + p'^2} \end{aligned}$$

or

$$\begin{aligned} \sqrt{m_1^2 c^2 + p'^2} \sqrt{m_2^2 c^2 + p'^2} &= m_2 E_{LAB} - p'^2 \\ \implies (m_1^2 c^2 + p'^2)(m_2^2 c^2 + p'^2) &= (m_2 E_{LAB} - p'^2)^2 \\ &= m_2^2 c^2 (m_1^2 c^2 + p_{LAB}^2) + p'^4 - 2m_2 p'^2 E_{LAB} \end{aligned}$$

where $E_{LAB}^2 = m_1^2 c^4 + p_{LAB}^2 c^2$

$$\begin{aligned} \implies [2m_2 E_{LAB} + (m_1^2 + m_2^2) c^2] p'^2 &= m_2^2 c^2 p_{LAB}^2 \\ \implies p' &= \frac{m_2 c^2 p_{LAB}}{W} \end{aligned}$$

What is the Velocity of the CM Frame relative to the Lab Frame in Relativity

For two particles in the lab frame, their energy-momentum 4-vectors are

$$\mathbf{p}_1 = (E_1/c, \vec{p}_1) = (\gamma_1 m_1 c, \vec{p}_1) \quad \text{and} \quad \mathbf{p}_2 = (E_2/c, \vec{p}_2) = (\gamma_2 m_2 c, \vec{p}_2)$$

and in the CM frame

$$\mathbf{p}'_1 = (E_1'/c, \vec{p}'_1) = (\gamma' m_1 c, \vec{p}') \quad \text{and} \quad \mathbf{p}'_2 = (E_2'/c, \vec{p}'_2) = (\gamma' m_2 c, -\vec{p}')$$

where $\gamma' = 1/\sqrt{1-v'^2/c^2}$. Let $\vec{\beta}_{CM}$ is the velocity between the lab and CM frame. We will use the Lorentz transformation to determine $\vec{\beta}_{CM}$. The Lorentz transformation of the momentum from the lab to CM frame is

$$\begin{cases} \vec{p}' &= \vec{p}_1 + \frac{\gamma_{CM}-1}{\beta_{CM}^2} (\vec{\beta}_{CM} \cdot \vec{p}_1) \vec{\beta}_{CM} - \gamma_{CM} \vec{\beta}_{CM} E_1/c \\ -\vec{p}' &= \vec{p}_2 + \frac{\gamma_{CM}-1}{\beta_{CM}^2} (\vec{\beta}_{CM} \cdot \vec{p}_2) \vec{\beta}_{CM} - \gamma_{CM} \vec{\beta}_{CM} E_2/c \end{cases}$$

$$\Rightarrow 0 = \vec{p}_{CM} + \frac{\gamma_{CM}-1}{\beta_{CM}^2} (\vec{\beta}_{CM} \cdot \vec{p}_{CM}) \vec{\beta}_{CM} - \gamma_{CM} \vec{\beta}_{CM} (E_1 + E_2)/c$$

where $\vec{p}_{CM}$ is the CM momentum observed in the lab frame,

$$\vec{p}_{CM} = \vec{p}_1 + \vec{p}_2$$

To solve $\vec{\beta}_{CM}$, we first determine its direction. Since

$$\begin{aligned} 0 &= \left[\vec{p}_{CM} + \frac{\gamma_{CM}-1}{\beta_{CM}^2} (\vec{\beta}_{CM} \cdot \vec{p}_{CM}) \vec{\beta}_{CM} - \gamma_{CM} \vec{\beta}_{CM} (E_1 + E_2)/c \right] \times \vec{\beta}_{CM} \\ &= \vec{p}_{CM} \times \vec{\beta}_{CM}, \end{aligned}$$

$\vec{\beta}_{CM}$ and $\vec{p}_{CM}$ are in the same direction, *i.e.*

$$\vec{p}_{CM} \cdot \vec{\beta}_{CM} = p_{CM} \beta_{CM}$$

and, therefore,

$$\begin{aligned} 0 &= \vec{p}_{CM} + \frac{\gamma_{CM}-1}{\beta_{CM}^2} (\vec{\beta}_{CM} \cdot \vec{p}_{CM}) \vec{\beta}_{CM} - \gamma_{CM} \vec{\beta}_{CM} (E_1 + E_2)/c \\ &= \vec{p}_{CM} + (\gamma_{CM} - 1) \vec{p}_{CM} - \gamma_{CM} \vec{\beta}_{CM} (E_1 + E_2)/c \\ &= \gamma_{CM} \vec{p}_{CM} - \gamma_{CM} \vec{\beta}_{CM} (E_1 + E_2)/c \\ &\Rightarrow c \vec{\beta}_{CM} = \frac{\vec{p}_{CM}}{(E_1 + E_2)/c^2} = \frac{\vec{p}_1 + \vec{p}_2}{\gamma_1 m_1 + \gamma_2 m_2} \end{aligned}$$

(a) (Continue) In this problem, the CM momentum is

$$\vec{p}_{CM} = \vec{p}_1 + \vec{p}_2 = \vec{p}_{LAB}$$

and the velocity the CM frame respect to the lab frame is

$$c \vec{\beta}_{CM} = \frac{c^2 \vec{p}_{LAB}}{E_1 + E_2} = \frac{\vec{p}_{LAB}}{\gamma_1 m_1 + m_2}$$

where $\gamma_2 = 1$ because $\vec{p}_2 = 0$. For particle 2, the Lorentz transformation of the momentum from the lab to CM frame is

$$-\vec{p}' = \vec{p}_2 + \frac{\gamma_{CM}-1}{\beta_{CM}^2} (\vec{\beta}_{CM} \cdot \vec{p}_2) \vec{\beta}_{CM} - \gamma_{CM} \vec{\beta}_{CM} E_2/c = -\gamma_{CM} \vec{\beta}_{CM} m_2 c$$

Thus

$$\vec{p}' = \frac{\gamma_{CM} m_2 c^2}{E_1 + E_2} \vec{p}_{LAB}$$

To transfer $E_1 + E_2$ to $W = E'_1 + E'_2$, we use the Lorentz transformation of 0-component of energy-momentum 4-vector,

$$\begin{aligned} E'_1/c &= \gamma_{CM}(E_1/c - \vec{\beta}_{CM} \cdot \vec{p}_1) \quad \text{and} \quad E'_2/c = \gamma_{CM}(E_2/c - \vec{\beta}_{CM} \cdot \vec{p}_2) \\ \Rightarrow \quad W &= E'_1 + E'_2 = \gamma_{CM} \left[(E_1 + E_2) - c \vec{\beta}_{CM} \cdot \vec{p}_{CM} \right] \\ &= \gamma_{CM} \left[(E_1 + E_2) - \beta_{CM}^2 (E_1 + E_2) \right] = (E_1 + E_2)/\gamma_{CM} \\ \Rightarrow \quad \gamma_{CM} &= \frac{E_1 + E_2}{W} = \frac{E_{LAB} + m_2 c^2}{W} \end{aligned}$$

and, therefore,

$$\vec{p}' = \frac{m_2 c^2}{W} \vec{p}_{LAB}$$

(b)

$$\vec{\beta}_{CM} = \frac{c \vec{p}_{LAB}}{E_1 + E_2} = \frac{c \vec{p}_{LAB}}{E_{LAB} + m_2 c^2}$$

(c)

When $\beta_1 \ll 1$ and $\beta_2 \ll 1$, $\gamma_1 \simeq 1$, $\gamma_2 \simeq 1$, and

$$\begin{aligned} \vec{\beta}_{CM} &= \frac{c \vec{p}_{LAB}}{E_1 + E_2} = \frac{c \vec{p}_{LAB}}{\lambda_1 m_1 c^2 + \lambda_2 m_2 c^2} = \frac{\vec{p}_{LAB}}{(m_1 + m_2)c} \ll 1 \\ \Rightarrow \quad \vec{v}_{CM} &= \frac{\vec{p}_1 + \vec{p}_2}{m_1 + m_2} \end{aligned}$$

and $\gamma_{CM} \simeq 1$,

$$\begin{aligned} \vec{p}' &= \frac{\gamma_{CM} m_2 c^2}{E_1 + E_2} \vec{p}_{LAB} \simeq \frac{m_2}{m_1 + m_2} \vec{p}_{LAB} \\ W &= (E_1 + E_2)/\gamma_{CM} = (E_1 + E_2) \sqrt{1 - \beta_{CM}^2} \\ &= (E_1 + E_2) \left(1 - \frac{1}{2} \beta_{CM}^2 \right) + O(p_{LAB}^4) \\ &= E_1 + E_2 - \frac{c^2 p_{LAB}^2}{2(E_1 + E_2)} + O(p_{LAB}^4) \\ &= m_1 c^2 + \frac{p_{LAB}^2}{2m_1} + m_2 c^2 - \frac{p_{LAB}^2}{2(m_1 + m_2)} + O(p_{LAB}^4) \\ &\simeq m_1 c^2 + m_2 c^2 + \frac{m_2 p_{LAB}^2}{2m_1(m_1 + m_2)} \end{aligned}$$

HW Assignment 11-4 (11.9, 11.10, 11.11)

11.9 An infinitesimal Lorentz transformation and its inverse can be written as

$$x'^{\alpha} = (g^{\alpha\beta} + \epsilon^{\alpha\beta})x_{\beta} \quad \text{and} \quad x^{\alpha} = (g^{\alpha\beta} + \epsilon'^{\alpha\beta})x'_{\beta}$$

where $\epsilon^{\alpha\beta}$ and $\epsilon'^{\alpha\beta}$ are infinitesimal.

(a) Since $x_{\beta} = g_{\beta\gamma}x^{\gamma}$ and

$$\begin{aligned} x'^{\alpha} &= (g^{\alpha\beta} + \epsilon^{\alpha\beta})x_{\beta} = (g^{\alpha\beta} + \epsilon^{\alpha\beta})g_{\beta\gamma}x^{\gamma} \\ &= (g^{\alpha\beta} + \epsilon^{\alpha\beta})g_{\beta\gamma}(g^{\gamma\zeta} + \epsilon'^{\gamma\zeta})x'_{\zeta} = (g^{\alpha\beta} + \epsilon^{\alpha\beta})g_{\beta\gamma}(g^{\gamma\zeta} + \epsilon'^{\gamma\zeta})g_{\zeta\xi}x'^{\xi}, \\ I^{\alpha\xi} &= (g^{\alpha\beta} + \epsilon^{\alpha\beta})g_{\beta\gamma}(g^{\gamma\zeta} + \epsilon'^{\gamma\zeta})g_{\zeta\xi}x'^{\xi} \end{aligned}$$

or

$$\begin{aligned} \mathbf{I} &= (\mathbf{g} + \boldsymbol{\epsilon})\mathbf{g}(\mathbf{g} + \boldsymbol{\epsilon}')\mathbf{g} = (\mathbf{I} + \boldsymbol{\epsilon}\mathbf{g})(\mathbf{I} + \boldsymbol{\epsilon}'\mathbf{g}) = \mathbf{I} + \boldsymbol{\epsilon}\mathbf{g} + \boldsymbol{\epsilon}'\mathbf{g} + O(\epsilon^2) \\ \implies \quad \boldsymbol{\epsilon}\mathbf{g} + \boldsymbol{\epsilon}'\mathbf{g} &= 0 \quad \implies \quad \boldsymbol{\epsilon}' = -\boldsymbol{\epsilon} \quad \text{or} \quad \epsilon'^{\alpha\beta} = -\epsilon^{\alpha\beta} \end{aligned}$$

(b) The preservation of norm is

$$\begin{aligned} x^{\alpha}x_{\alpha} &= x'^{\alpha}x'_{\alpha} = (g^{\alpha\beta} + \epsilon^{\alpha\beta})x_{\beta}g_{\alpha\gamma}(g^{\gamma\zeta} + \epsilon'^{\gamma\zeta})x_{\zeta} \\ &= (g^{\alpha\beta} + \epsilon^{\alpha\beta})g_{\alpha\gamma}(g^{\gamma\zeta} + \epsilon'^{\gamma\zeta})g_{\zeta\xi}x^{\xi}x_{\beta} \end{aligned}$$

Thus,

$$I_{\xi\beta} = (g^{\alpha\beta} + \epsilon^{\alpha\beta})g_{\alpha\gamma}(g^{\gamma\zeta} + \epsilon'^{\gamma\zeta})g_{\zeta\xi} = [(g^{\gamma\zeta} + \epsilon'^{\gamma\zeta})g_{\zeta\xi}][g_{\gamma\alpha}(g^{\alpha\beta} + \epsilon^{\alpha\beta})]$$

or

$$\begin{aligned} \mathbf{I} &= [(\mathbf{g} + \boldsymbol{\epsilon})\mathbf{g}]^T [\mathbf{g}(\mathbf{g} + \boldsymbol{\epsilon})] = (\mathbf{I} + \boldsymbol{\epsilon}\mathbf{g})^T (\mathbf{I} + \mathbf{g}\boldsymbol{\epsilon}) = \mathbf{I} + \mathbf{g}\boldsymbol{\epsilon}^T + \mathbf{g}\boldsymbol{\epsilon} + O(\epsilon^2) \\ \implies \quad \boldsymbol{\epsilon}^T &= -\boldsymbol{\epsilon} \quad \text{or} \quad \epsilon^{\beta\alpha} = -\epsilon^{\alpha\beta} \end{aligned}$$

(c)

$$x'^{\alpha} = (g^{\alpha\beta} + \epsilon^{\alpha\beta})x_{\beta} = (g^{\alpha\beta} + \epsilon^{\alpha\beta})g_{\beta\gamma}x^{\gamma}$$

or

$$\mathbf{x}' = (\mathbf{g} + \boldsymbol{\epsilon})\mathbf{g}\mathbf{x} = (\mathbf{I} + \boldsymbol{\epsilon}\mathbf{g})\mathbf{x}$$

On the other hand, from the infinitesimal generator of Lorentz transformation,

$$\mathbf{x}' = e^{\mathbf{L}}\mathbf{x} = \sum_{n=0}^{\infty} \frac{1}{n!} \mathbf{L}^n$$

For infinitesimal $\mathbf{L}$, the expansion can be truncated as

$$\mathbf{x}' = e^{\mathbf{L}}\mathbf{x} = (\mathbf{I} + \mathbf{L})\mathbf{x}$$

Therefore,

$$\boldsymbol{\epsilon}\mathbf{g} = \mathbf{L} \quad \text{or} \quad \boldsymbol{\epsilon} = \mathbf{L}\mathbf{g}$$

11.10 (a) Since

$$\vec{\epsilon} \cdot \vec{S} = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -\epsilon_3 & \epsilon_2 \\ 0 & \epsilon_3 & 0 & -\epsilon_1 \\ 0 & -\epsilon_2 & \epsilon_1 & 0 \end{bmatrix} \quad \text{and} \quad (\vec{\epsilon} \cdot \vec{S})^2 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & & & \\ 0 & \delta_{ij} + \epsilon_i \epsilon_j & & \\ 0 & & & \end{bmatrix},$$

$$(\vec{\epsilon} \cdot \vec{S})^3 = \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -\epsilon_3 & \epsilon_2 \\ 0 & \epsilon_3 & 0 & -\epsilon_1 \\ 0 & -\epsilon_2 & \epsilon_1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -\epsilon_3 & \epsilon_2 \\ 0 & \epsilon_3 & 0 & -\epsilon_1 \\ 0 & -\epsilon_2 & \epsilon_1 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -\epsilon_3 & \epsilon_2 \\ 0 & \epsilon_3 & 0 & -\epsilon_1 \\ 0 & -\epsilon_2 & \epsilon_1 & 0 \end{bmatrix} = -\vec{\epsilon} \cdot \vec{S}$$

where $\vec{\epsilon} \cdot \vec{\epsilon} = 1$. Since

$$\vec{\epsilon} \cdot \vec{K} = \begin{bmatrix} 0 & \epsilon_1 & \epsilon_2 & \epsilon_3 \\ \epsilon_1 & 0 & 0 & 0 \\ \epsilon_2 & 0 & 0 & 0 \\ \epsilon_3 & 0 & 0 & 0 \end{bmatrix} \quad \text{and} \quad (\vec{\epsilon} \cdot \vec{K})^2 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \epsilon_1^2 & \epsilon_1 \epsilon_2 & \epsilon_1 \epsilon_3 \\ 0 & \epsilon_2 \epsilon_1 & \epsilon_2^2 & \epsilon_2 \epsilon_3 \\ 0 & \epsilon_3 \epsilon_1 & \epsilon_2 \epsilon_2 & \epsilon_3^2 \end{bmatrix},$$

$$(\vec{\epsilon} \cdot \vec{K})^3 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \epsilon_1^2 & \epsilon_1 \epsilon_2 & \epsilon_1 \epsilon_3 \\ 0 & \epsilon_2 \epsilon_1 & \epsilon_2^2 & \epsilon_2 \epsilon_3 \\ 0 & \epsilon_3 \epsilon_1 & \epsilon_2 \epsilon_2 & \epsilon_3^2 \end{bmatrix} \begin{bmatrix} 0 & \epsilon_1 & \epsilon_2 & \epsilon_3 \\ \epsilon_1 & 0 & 0 & 0 \\ \epsilon_2 & 0 & 0 & 0 \\ \epsilon_3 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & \epsilon_1 & \epsilon_2 & \epsilon_3 \\ \epsilon_1 & 0 & 0 & 0 \\ \epsilon_2 & 0 & 0 & 0 \\ \epsilon_3 & 0 & 0 & 0 \end{bmatrix} = \vec{\epsilon} \cdot \vec{K}$$

11.11 Following the hint, we expand $A(\lambda) = e^{\lambda(L+\delta L)} e^{-\lambda L}$ into a Taylor series of λ with a small δL ($\delta L \ll I$).

$$\begin{aligned} A(0) &= I \\ \frac{dA}{d\lambda} &= e^{\lambda(L+\delta L)} (L + \delta L) e^{-\lambda L} - e^{\lambda(L+\delta L)} L e^{-\lambda L} \\ &= e^{\lambda(L+\delta L)} (L + \delta L - L) e^{-\lambda L} = e^{\lambda(L+\delta L)} \delta L e^{-\lambda L} \\ \frac{d^2 A}{d\lambda^2} &= e^{\lambda(L+\delta L)} [(L + \delta L)\delta L - \delta L L] e^{-\lambda L} \\ &= e^{\lambda(L+\delta L)} \{[L, \delta L] + (\delta L)^2\} e^{-\lambda L} = e^{\lambda(L+\delta L)} [L, \delta L] e^{-\lambda L} + O(\delta L^2) \\ \frac{d^3 A}{d\lambda^3} &= e^{\lambda(L+\delta L)} \{(L + \delta L)[L, \delta L] - [L, \delta L]L\} e^{-\lambda L} + O(\delta L^2) \\ &= e^{\lambda(L+\delta L)} [L, [L, \delta L]] e^{-\lambda L} + O(\delta L^2) \\ \frac{d^4 A}{d\lambda^4} &= e^{\lambda(L+\delta L)} \{(L + \delta L)[L, [L, \delta L]] - [L, [L, \delta L]]L\} e^{-\lambda L} + O(\delta L^2) \\ &= e^{\lambda(L+\delta L)} [L, [L, [L, \delta L]]] e^{-\lambda L} + O(\delta L^2) \end{aligned}$$

Neglecting $O(\delta L^2)$, the Taylor expansion of $A(\lambda)$ is

$$\begin{aligned} A(\lambda) &= \sum_{n=0}^{\infty} \frac{1}{n!} \left. \frac{d^n A}{d\lambda^n} \right|_{\lambda=0} \lambda^n \\ &= I + \delta L \lambda + \frac{1}{2!} [L, \delta L] \lambda^2 + \frac{1}{3!} [L, [L, \delta L]] \lambda^3 + \frac{1}{4!} [L, [L, [L, \delta L]]] \lambda^4 + \dots \end{aligned}$$

Setting $\lambda = 1$ yields

$$e^{L+\delta L} e^{-L} = I + \delta L + \frac{1}{2!} [L, \delta L] + \frac{1}{3!} [L, [L, \delta L]] + \frac{1}{4!} [L, [L, [L, \delta L]]] + \dots$$

HW Assignment 11-5 (11.13, 11.14, Extra 5)

11.13 An infinitely long straight wire with a uniform linear charge density ρ_0 is at rest in the initial frame K' .

(a) Electric and Magnetic Field in the Rest and Lab Frame

In the rest frame K' , all the charges are static and, therefore, $\vec{B}' = 0$. With a cylindrical coordinate (r', θ', z') in the K' frame with z -axis along the wire, the electric field can be easily calculated using the Gaussian law,

$$\vec{E}' = \rho_0 \frac{\vec{r}'}{r'^2} = E' \vec{e}_{r'} \quad \text{where} \quad E' = \frac{\rho_0}{r'}$$

Note that in the 2-dimensional space, the electric field of a point charge is $\sim 1/r'$.

In the lab frame, K' is moving with a velocity v in the direction parallel to the wire, *i.e.* $\vec{\beta} \perp \vec{e}_{r'}$. The Lorentz transformation of the field is

$$\begin{aligned} \vec{E} &= \gamma(\vec{E}' - \vec{\beta} \times \vec{B}') - \frac{\gamma^2}{\gamma+1} \vec{\beta}(\vec{\beta} \cdot \vec{E}') = \gamma E' \vec{e}_r \\ \vec{B} &= \gamma(\vec{B}' + \vec{\beta} \times \vec{E}') - \frac{\gamma^2}{\gamma+1} \vec{\beta}(\vec{\beta} \cdot \vec{B}') = \gamma \beta E' \vec{e}_z \times \vec{e}_r = \gamma \beta E' \vec{e}_\theta \end{aligned}$$

where

$$E' = \frac{\rho_0}{r'} = \frac{\rho_0}{r} \quad \text{with} \quad r = r'$$

since $\vec{e}_r = \vec{e}_{r'}$ is perpendicular to $\vec{\beta}$.

(b) Charge and current density in the rest and lab frames.

In the rest frame K' , the charge and current density are

$$\rho' = \rho_0 \quad \text{and} \quad \vec{J}' = 0$$

The 4-vector for the charge/current density is thus

$$\mathbf{J}' = (c\rho_0, 0, 0, 0)$$

where J'^3 is in the direction of $\vec{\beta}$. The Lorentz transformation of $\mathbf{J}'$ to the lab frame is

$$\begin{cases} J^0 = \gamma(J'^0 + \beta J'^3) = \gamma J'^0 \\ J^3 = \gamma(J'^3 + \beta J'^0) = \gamma\beta J'^0 \\ J^1 = J'^1 \\ J^2 = J'^2 \end{cases}$$

In the lab frame, therefore, the charge density and current are

$$\rho = \gamma\rho_0 \quad \text{and} \quad \vec{J} = \rho\vec{v}$$

The increase of charge density observed in the lab can be understood as the length contraction of the wire (in the direction of motion) while the number of charges are the same in the lab and rest frame.

(c) In the lab frame, for a long current carrying wire, the electric and magnetic field can directly be calculated using the Gaussian and Ampere law, respectively, with the charge and current density in the lab frame as

$$\vec{E} = \frac{\rho}{r} \vec{e}_r = \frac{\gamma\rho_0}{r} \vec{e}_r \quad \text{and} \quad \vec{B} = \frac{\vec{J} \times \vec{e}_r}{cr} = \frac{\gamma\rho_0\beta}{r} \vec{e}_\theta$$

which is the same result of Part (a).

11.14 (a)

$$F^{\alpha\beta} = \begin{pmatrix} 0 & -E_1 & -E_2 & -E_3 \\ E_1 & 0 & -B_3 & B_2 \\ E_2 & B_3 & 0 & -B_1 \\ E_3 & -B_2 & B_1 & 0 \end{pmatrix} \quad \text{and} \quad F_{\alpha\beta} = \begin{pmatrix} 0 & E_1 & E_2 & E_3 \\ -E_1 & 0 & -B_3 & B_2 \\ -E_2 & B_3 & 0 & -B_1 \\ -E_3 & -B_2 & B_1 & 0 \end{pmatrix}$$

where $F_{\alpha\beta} = -F_{\beta\alpha}$ and $F^{\alpha\beta} = -F^{\beta\alpha}$.

$$\mathcal{F}^{\alpha\beta} = \begin{pmatrix} 0 & -B_1 & -B_2 & -B_3 \\ B_1 & 0 & E_3 & -E_2 \\ B_2 & -E_3 & 0 & E_1 \\ B_3 & E_2 & -E_1 & 0 \end{pmatrix} \quad \text{and} \quad \mathcal{F}_{\alpha\beta} = \begin{pmatrix} 0 & B_1 & B_2 & B_3 \\ -B_1 & 0 & E_3 & -E_2 \\ -B_2 & -E_3 & 0 & E_1 \\ -B_3 & E_2 & -E_1 & 0 \end{pmatrix}$$

where $\mathcal{F}_{\alpha\beta} = -\mathcal{F}_{\beta\alpha}$ and $\mathcal{F}^{\alpha\beta} = -\mathcal{F}^{\beta\alpha}$.

$$\begin{aligned}
F^{\alpha\beta}F_{\alpha\beta} &= -F^{\alpha\beta}F_{\beta\alpha} = -\sum_{\alpha}\sum_{\beta}F^{\alpha\beta}F_{\beta\alpha} \\
&= \text{Tr} \left[\begin{pmatrix} 0 & -E_1 & -E_2 & -E_3 \\ E_1 & 0 & -B_3 & B_2 \\ E_2 & B_3 & 0 & -B_1 \\ E_3 & -B_2 & B_1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -E_1 & -E_2 & -E_3 \\ E_1 & 0 & B_3 & -B_2 \\ E_2 & -B_3 & 0 & B_1 \\ E_3 & B_2 & -B_1 & 0 \end{pmatrix} \right] \\
&= -(E_1^2 + E_2^2 + E_3^2) + (B_3^2 + B_2^2 - E_1^2) + (B_3^2 + B_1^2 - E_2^2) + (B_2^2 + B_1^2 - E_3^2) \\
&= 2(\vec{B} \cdot \vec{B} - \vec{E} \cdot \vec{E})
\end{aligned}$$

$$\begin{aligned}
\mathcal{F}^{\alpha\beta}F_{\alpha\beta} &= -\mathcal{F}^{\alpha\beta}F_{\beta\alpha} = -\sum_{\alpha}\sum_{\beta}\mathcal{F}^{\alpha\beta}F_{\beta\alpha} \\
&= \text{Tr} \left[\begin{pmatrix} 0 & -B_1 & -B_2 & -B_3 \\ B_1 & 0 & E_3 & -E_2 \\ B_2 & -E_3 & 0 & E_1 \\ B_3 & E_2 & -E_1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -E_1 & -E_2 & -E_3 \\ E_1 & 0 & B_3 & -B_2 \\ E_2 & -B_3 & 0 & B_1 \\ E_3 & B_2 & -B_1 & 0 \end{pmatrix} \right] \\
&= -4\vec{B} \cdot \vec{E} \\
\mathcal{F}^{\alpha\beta}\mathcal{F}_{\alpha\beta} &= -\mathcal{F}^{\alpha\beta}F_{\beta\alpha} = -\sum_{\alpha}\sum_{\beta}\mathcal{F}^{\alpha\beta}F_{\beta\alpha} \\
&= \text{Tr} \left[\begin{pmatrix} 0 & -B_1 & -B_2 & -B_3 \\ B_1 & 0 & E_3 & -E_2 \\ B_2 & -E_3 & 0 & E_1 \\ B_3 & E_2 & -E_1 & 0 \end{pmatrix} \begin{pmatrix} 0 & -B_1 & -B_2 & -B_3 \\ B_1 & 0 & -E_3 & E_2 \\ B_2 & E_3 & 0 & -E_1 \\ B_3 & -E_2 & E_1 & 0 \end{pmatrix} \right] \\
&= 2(\vec{E} \cdot \vec{E} - \vec{B} \cdot \vec{B})
\end{aligned}$$

$(\vec{B} \cdot \vec{E})$ and $(\vec{E} \cdot \vec{E} - \vec{B} \cdot \vec{B})$ are only two invariants that are quadratic in the field strengths $\vec{E}$ and $\vec{B}$.

11.14 (b) No, it is not possible to have an electromagnetic field that appears as a purely electric field in one inertial frame and as a purely magnetic field in some other inertial frames since

$$E^2 - B^2 = \text{constant}$$

and if $B^2 = 0$ and $E^2 - B^2 = E^2 = \text{constant} > 0$ in one inertial frame, it is not possible to have $E^2 = 0$ with $E^2 - B^2 = -B^2 = \text{constant} < 0$ in any other inertial frame.

Extra HW 5.

Is it possible to have $\vec{E}' \perp \vec{B}'$ in an inertial frame and $\vec{E} \cdot \vec{B} \neq 0$ in another inertial frame?

In the case of $\vec{\beta} \parallel (\vec{E} \times \vec{B})$,

$$\begin{aligned}\vec{E}' \cdot \vec{B}' &= \gamma^2(\vec{E} + \vec{\beta} \times \vec{B}) \cdot (\vec{B} - \vec{\beta} \times \vec{E}) = \gamma^2 [\vec{E} \cdot \vec{B} - (\vec{\beta} \times \vec{B}) \cdot (\vec{\beta} \times \vec{E})] \\ &= \gamma^2(\vec{E} \cdot \vec{B})(1 - \beta^2) = \vec{E} \cdot \vec{B}\end{aligned}$$

where $1 - \beta^2 = \gamma^{-2}$. Thus, if $\vec{E} \cdot \vec{B} \neq 0$, it is not possible to have $\vec{E}' \cdot \vec{B}' = 0$ in a moving frame with $\vec{\beta} \parallel (\vec{E} \times \vec{B})$.

With an arbitrary direction of $\vec{\beta}$,

$$\begin{aligned}\vec{E}' \cdot \vec{B}' &= \left\{ \gamma(\vec{E} + \vec{\beta} \times \vec{B}) - \frac{\gamma^2}{\gamma+1} \vec{\beta}(\vec{\beta} \cdot \vec{E}) \right\} \cdot \left\{ \gamma(\vec{B} - \vec{\beta} \times \vec{E}) - \frac{\gamma^2}{\gamma+1} \vec{\beta}(\vec{\beta} \cdot \vec{B}) \right\} \\ &= \gamma^2(\vec{E} + \vec{\beta} \times \vec{B}) \cdot (\vec{B} - \vec{\beta} \times \vec{E}) + \frac{\gamma^4 \beta^2}{(\gamma+1)^2} (\vec{\beta} \cdot \vec{B})(\vec{\beta} \cdot \vec{E}) \\ &\quad - \frac{\gamma^3}{\gamma+1} [(\vec{E} + \vec{\beta} \times \vec{B}) \cdot \vec{\beta}(\vec{\beta} \cdot \vec{B}) + (\vec{B} - \vec{\beta} \times \vec{E}) \cdot \vec{\beta}(\vec{\beta} \cdot \vec{E})] \\ &= \gamma^2 \left\{ \vec{E} \cdot \vec{B} - (\vec{\beta} \times \vec{B}) \cdot (\vec{\beta} \times \vec{E}) + \left[\frac{\gamma^2 \beta^2}{(\gamma+1)^2} - \frac{2\gamma}{\gamma+1} \right] (\vec{\beta} \cdot \vec{B})(\vec{\beta} \cdot \vec{E}) \right\} \\ &= \gamma^2 \left\{ \vec{E} \cdot \vec{B}(1 - \beta^2) + \left[\frac{\gamma^2 \beta^2}{(\gamma+1)^2} - \frac{2\gamma}{\gamma+1} + 1 \right] (\vec{\beta} \cdot \vec{B})(\vec{\beta} \cdot \vec{E}) \right\} \\ &= \vec{E} \cdot \vec{B}\end{aligned}$$

where $\beta^2 = (\gamma^2 - 1)/\gamma^2$ has been used. Therefore, $\vec{E} \cdot \vec{B}$ is invariant under Lorentz transformation and $\vec{E}'$ and $\vec{B}'$ cannot be made perpendicular to each other if both of them are nonzero and have an angle θ between them, where $\pi/2 > \theta > 0$ or $\pi > \theta > \pi/2$.

HW Assignment 11-6 (11.15, Extra 6)

11.15 In K frame,

$$\vec{E} = E_0 \vec{e}_x \quad \text{and} \quad \vec{B} = 2E_0(\cos \theta \vec{e}_x + \sin \theta \vec{e}_y)$$

We can choose K' frame is moving in $\vec{e}_z$ with respect to K frame, *i.e.* $\vec{\beta} = \beta \vec{e}_z$ and

$$\begin{aligned}\vec{E}' &= \gamma(\vec{E} + \vec{\beta} \times \vec{B}) - \frac{\gamma^2}{\gamma+1} \vec{\beta}(\vec{\beta} \cdot \vec{E}) = \gamma(\vec{E} + \vec{\beta} \times \vec{B}) \\ \vec{B}' &= \gamma(\vec{B} - \vec{\beta} \times \vec{E}) - \frac{\gamma^2}{\gamma+1} \vec{\beta}(\vec{\beta} \cdot \vec{B}) = \gamma(\vec{B} - \vec{\beta} \times \vec{E})\end{aligned}$$

In K' frame $\vec{E}' \parallel \vec{B}'$, $\vec{E}' \times \vec{B}' = 0$, i.e.

$$\begin{aligned}
 0 &= (\vec{E} + \vec{\beta} \times \vec{B}) \times (\vec{B} - \vec{\beta} \times \vec{E}) \\
 &= \vec{E} \times \vec{B} - \vec{E} \times (\vec{\beta} \times \vec{E}) - \vec{B} \times (\vec{\beta} \times \vec{B}) - (\vec{\beta} \times \vec{B}) \times (\vec{\beta} \times \vec{E}) \\
 &= \vec{E} \times \vec{B} - \vec{\beta}(E^2 + B^2) - \vec{\beta}[\vec{\beta} \cdot (\vec{B} \times \vec{E})] \\
 &\implies EB \sin \theta - \beta(E^2 + B^2) + \beta^2 EB \sin \theta = 0
 \end{aligned}$$

For arbitrary field strengths, therefore, $\vec{E}$ and $\vec{B}$ will be parallel in K' frame if

$$\beta = \frac{(E^2 + B^2) - \sqrt{(E^2 + B^2)^2 - 4(EB)^2 \sin^2 \theta}}{2EB \sin \theta} = \frac{1 - \sqrt{1 - \eta^2 \sin^2 \theta}}{\eta \sin \theta}$$

where

$$\eta = \frac{2EB}{E^2 + B^2} < 1$$

Note that for $\pi/2 < \theta < \pi$, this result is still valid with $\beta \rightarrow -\beta$ which is $\vec{\beta} = -|\beta|\vec{e}_z$. For $B = 2E$, $\eta = 4/5$ and

$$\beta = \frac{5 - \sqrt{25 - 16 \sin^2 \theta}}{4 \sin \theta} \quad \text{or} \quad \sin \theta = \frac{5\beta}{2(1 + \beta^2)}$$

When $\theta = 0$, $\beta = 0$. When $\theta = \pi/2$, $\beta = B/E$ if $B < E$ and $\beta = E/B$ if $B > E$.

Extra HW 6. Electromagnetic Force Between Ultra-Relativistic Electrons

Calculate the force between two ultra-relativistic electrons moving in (a) the same direction and (b) the opposite direction.

In the lab frame, the coordinate and velocity of the two electrons (electron 1 and 2) are

$$\begin{cases} \vec{r}_1 = (x_1, y_1, z_1) \\ \vec{r}_2 = (x_2, y_2, z_2) \end{cases} \quad \text{and} \quad \begin{cases} \vec{v}_1 = v_1 \vec{e}_x \\ \vec{v}_2 = v_2 \vec{e}_x \end{cases}$$

where $v_1 = v_2$ if they are moving in the same direction and $v_1 = -v_2$ if they are moving in the opposite direction. In the following, we will calculate the force on electron 1 exerted by electron 2.

To calculate the electric and magnetic field at the location of electron 1 due to electron 2, we first go to the rest frame of electron 2 and the field is simply

$$\vec{E}' = e \frac{\vec{r}'_2 - \vec{r}'_1}{|\vec{r}'_2 - \vec{r}'_1|^3} \quad \text{and} \quad \vec{B}' = 0$$

In the lab frame, from the Lorentz transformation,

$$\begin{cases} x'_2 - x'_1 &= \gamma_2(x_2 - x_1) \\ y'_2 - y'_1 &= y_2 - y_1 \\ z'_2 - z'_1 &= z_2 - z_1 \end{cases}$$

because $t'_1 = t'_2$ and

$$\begin{aligned} |\vec{r}'_2 - \vec{r}'_1| &= [(x'_2 - x'_1)^2 + (y'_2 - y'_1)^2 + (z'_2 - z'_1)^2]^{1/2} \\ &= [\gamma_2^2(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2]^{1/2} \end{aligned}$$

where $\gamma_2 = 1/\sqrt{1 - v_2^2/c^2}$. The field seen by electron 1 in the lab frame is

$$\begin{cases} \vec{E} &= \gamma_2 \vec{E}' - \frac{\gamma_2^2}{\gamma_2 + 1} \vec{\beta}_2 (\vec{\beta}_2 \cdot \vec{E}') \\ \vec{B} &= \gamma_2 (\vec{\beta}_2 \times \vec{E}') \end{cases}$$

where $\vec{\beta}_2 = \vec{v}_2/c$. For ultra-relativistic particles, $\beta_2 \simeq 1$, $\gamma_2 \gg 1$, and

$$\begin{cases} \vec{E} &= \gamma_2 [\vec{E}' - \vec{\beta}_2 (\vec{\beta}_2 \cdot \vec{E}')] = \gamma_2 \vec{E}'_{\perp} \\ \vec{B} &= \gamma_2 (\vec{\beta}_2 \times \vec{E}') = (-\beta_2 E_z) \vec{e}_y + (\beta_2 E_y) \vec{e}_z \end{cases}$$

where

$$\vec{E}'_{\perp} = \frac{e}{[\gamma_2^2(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2]^{3/2}} [(y_2 - y_1) \vec{e}_y + (z_2 - z_1) \vec{e}_z]$$

In the lab frame, the Lorentz force on electron 1 is

$$\vec{F} = e(\vec{E} + \vec{\beta}_1 \times \vec{B})$$

where $\vec{\beta}_1 = \vec{v}_1/c$ and

$$\vec{\beta}_1 \times \vec{B} = \gamma_2 \vec{\beta}_1 \times (\vec{\beta}_2 \times \vec{E}') = \gamma_2 [(\vec{\beta}_1 \cdot \vec{E}') \vec{\beta}_2 - (\vec{\beta}_1 \cdot \vec{\beta}_2) \vec{E}'] = \begin{cases} -\vec{E}, & \text{if } \vec{\beta}_1 = \vec{\beta}_2 \\ \vec{E}, & \text{if } \vec{\beta}_1 = -\vec{\beta}_2 \end{cases}$$

Therefore,

$$\vec{F} = \begin{cases} 0, & \text{if } \vec{\beta}_1 = \vec{\beta}_2 \\ 2e\vec{E}, & \text{if } \vec{\beta}_1 = -\vec{\beta}_2 \end{cases}$$

HW Assignment 11-7 (Chapter 11) (Extra 7, Extra 9)

Extra HW 7. Consider a uniform charge distribution on a spherical shell of radius r_0 and mass m . The spherical shell rotates with an angular velocity $\vec{\omega}$.

(a) Calculate the magnetic dipole moment of the shell.

For convenience, we can choose a coordinate with the z -axis parallel to the direction of $\vec{\omega}$. The linear velocity on the shell along the tangential direction of the rotation is then

$$\vec{v} = \vec{\omega} \times \vec{r}$$

In the spherical coordinate, the current density on the spherical shell can be written as

$$\vec{J}(r, \phi, \theta) = \rho_0 \vec{v} \delta(r - r_0) = \rho_0 (\vec{\omega} \times \vec{r}) \delta(r - r_0) = \frac{q}{4\pi r_0^2} (\vec{\omega} \times \vec{r}) \delta(r - r_0)$$

where $\rho_0 = q/(4\pi r_0^2)$ is the charge density on the shell. The magnetic dipole moment is

$$\vec{m} = \frac{1}{2c} \int [\vec{r} \times \vec{J}(\vec{r})] d\vec{r} = \frac{q}{8\pi c r_0} \int_0^\pi \int_0^{2\pi} [\vec{r}_0 \times (\vec{\omega} \times \vec{r}_0)] \sin \theta d\theta d\phi$$

where

$$\vec{r}_0 \times (\vec{\omega} \times \vec{r}_0) = (\vec{r}_0 \cdot \vec{r}_0) \vec{\omega} - (\vec{r}_0 \cdot \vec{\omega}) \vec{r}_0 = r_0^2 \vec{\omega} - (r_0 \omega \cos \theta) \vec{r}_0$$

$$m_x \sim \int [\vec{r}_0 \times (\vec{\omega} \times \vec{r}_0)]_x \sin \theta d\theta d\phi = - \int \cos \phi \sin^2 \theta \cos \theta d\theta d\phi = 0$$

$$m_y \sim \int [\vec{r}_0 \times (\vec{\omega} \times \vec{r}_0)]_y \sin \theta d\theta d\phi = - \int \sin \phi \sin^2 \theta \cos \theta d\theta d\phi = 0$$

$$m_z = \frac{q}{8\pi c r_0} \int [\vec{r}_0 \times (\vec{\omega} \times \vec{r}_0)]_z \sin \theta d\theta d\phi = \frac{q r_0 \omega}{8\pi c} \int (1 - \cos^2 \phi) \sin \theta d\theta d\phi = \frac{q r_0 \omega}{4c}$$

where

$$\int_0^\pi \int_0^{2\pi} (1 - \cos^2 \phi) \sin \theta d\theta d\phi = 2\pi$$

Therefore

$$\vec{m} = \frac{q}{4c} r_0 \vec{\omega}$$

(b)

The mass density of the shell

$$dm = \frac{m}{4\pi r_0^2} \delta(r - r_0) r^2 \sin \theta d\theta d\phi dr$$

The angular momentum of the rotating shell is

$$\vec{L} = \int (\vec{r} \times \vec{v}) dm = \frac{m}{4\pi} \int_0^\pi \int_0^{2\pi} [\vec{r}_0 \times (\vec{\omega} \times \vec{r}_0)] \sin \theta d\theta d\phi = \frac{m}{4\pi} \frac{8\pi c r_0 \vec{m}}{q} = \frac{2m c r_0}{q} \vec{m}$$

Extra HW 9.

(a) Derive the equation of motion of the longitudinal component of a spin when both electric and magnetic field are included,

$$\frac{ds_{\parallel}}{dt} = \frac{q\gamma}{mc} \left[\left(\frac{g}{2} - 1 \right) \vec{s}_{\perp} \cdot (\vec{B} \times \hat{\beta}) + \left(\frac{1}{\beta} - \frac{g\beta}{2} \right) \vec{s}_{\perp} \cdot \vec{E} \right]$$

With both $\vec{E}$ and $\vec{B}$ field, the space components of BMT equation are

$$\frac{d\vec{s}}{dt} = \frac{gq}{2\gamma mc} \left[(\vec{\beta} \cdot \vec{s}) \vec{E} + \vec{s} \times \vec{B} \right] + \frac{\gamma(2-g)q}{2mc} \left[\vec{s} \cdot (\vec{E} + \vec{\beta} \times \vec{B}) - (\vec{\beta} \cdot \vec{s})(\vec{\beta} \cdot \vec{E}) \right] \vec{\beta} \quad (7)$$

The equation for the longitudinal component of a spin can be obtained from

$$\frac{ds_{\parallel}}{dt} = \frac{d}{dt}(\vec{s} \cdot \hat{\beta}) = \frac{d\vec{s}}{dt} \cdot \hat{\beta} + \vec{s} \cdot \frac{d\hat{\beta}}{dt} \quad \text{where} \quad \hat{\beta} = \frac{\vec{\beta}}{\beta} \quad (8)$$

Since the equation of the orbital motion is

$$\frac{d\gamma}{dt} = \frac{q}{mc} (\vec{\beta} \cdot \vec{E}) \quad \text{and} \quad \frac{d\vec{\beta}}{dt} = \frac{q}{\gamma mc} [\vec{E} + \vec{\beta} \times \vec{B} - (\vec{\beta} \cdot \vec{E})\vec{\beta}]$$

and also

$$\frac{d\gamma}{dt} = \frac{\beta}{(1 - \beta^2)^{3/2}} \frac{d\beta}{dt} \quad \Rightarrow \quad \frac{d\beta}{dt} = \frac{1}{\beta\gamma^3} \frac{d\gamma}{dt} = \frac{q}{\beta\gamma^3 mc} (\vec{\beta} \cdot \vec{E})$$

where $\gamma = 1/\sqrt{1 - \beta^2}$,

$$\begin{aligned} \frac{d\hat{\beta}}{dt} &= \frac{1}{\beta} \frac{d\vec{\beta}}{dt} - \frac{\vec{\beta}}{\beta^2} \frac{d\beta}{dt} \\ &= \frac{q}{\beta\gamma mc} [\vec{E} + \vec{\beta} \times \vec{B} - (\vec{\beta} \cdot \vec{E})\vec{\beta}] - \frac{q}{\beta^3\gamma^3 mc} (\vec{\beta} \cdot \vec{E}) \vec{\beta} \\ &= \frac{q}{\beta\gamma mc} \left[\vec{E} + \vec{\beta} \times \vec{B} - \frac{1}{\beta^2} (\vec{\beta} \cdot \vec{E}) \vec{\beta} \right] \end{aligned}$$

Substituting $d\vec{s}/dt$ from BMT equation and $d\hat{\beta}/dt$ into Eq. (8) yields

$$\begin{aligned} \frac{ds_{\parallel}}{dt} &= \frac{1}{\beta} \left(\frac{d\vec{s}}{dt} \cdot \vec{\beta} \right) + \vec{s} \cdot \frac{d\hat{\beta}}{dt} \\ &= \frac{gq}{2\beta\gamma mc} [(\vec{\beta} \cdot \vec{s})(\vec{\beta} \cdot \vec{E}) + \vec{\beta} \cdot (\vec{s} \times \vec{B})] + \frac{\beta\gamma(2-g)q}{2mc} [\vec{s} \cdot (\vec{E} + \vec{\beta} \times \vec{B}) - (\vec{\beta} \cdot \vec{s})(\vec{\beta} \cdot \vec{E})] \\ &\quad + \frac{q}{\beta\gamma mc} \left[(\vec{s} \cdot \vec{E}) + \vec{s} \cdot (\vec{\beta} \times \vec{B}) - \frac{1}{\beta^2} (\vec{\beta} \cdot \vec{E})(\vec{s} \cdot \vec{\beta}) \right] \\ &= \frac{q}{2\beta\gamma mc} \left\{ g(\vec{\beta} \cdot \vec{s})(\vec{\beta} \cdot \vec{E}) + g\vec{\beta} \cdot (\vec{s} \times \vec{B}) + \beta^2\gamma^2(2-g) [\vec{s} \cdot (\vec{E} + \vec{\beta} \times \vec{B}) - (\vec{\beta} \cdot \vec{s})(\vec{\beta} \cdot \vec{E})] \right. \\ &\quad \left. + 2(\vec{s} \cdot \vec{E}) + 2\vec{s} \cdot (\vec{\beta} \times \vec{B}) - \frac{2}{\beta^2} (\vec{\beta} \cdot \vec{s})(\vec{\beta} \cdot \vec{E}) \right\} \\ &= \frac{q}{2\beta\gamma mc} \left\{ \left[g - \beta^2\gamma^2(2-g) - \frac{2}{\beta^2} \right] (\vec{\beta} \cdot \vec{s})(\vec{\beta} \cdot \vec{E}) + [2 + \beta^2\gamma^2(2-g) - g] \vec{s} \cdot (\vec{\beta} \times \vec{B}) \right. \\ &\quad \left. + [2 + \beta^2\gamma^2(2-g)] (\vec{s} \cdot \vec{E}) \right\} \\ &= \frac{q}{2\beta\gamma mc} \left\{ \gamma^2(2-g)[\vec{s} \cdot (\vec{\beta} \times \vec{B})] - \left[\gamma^2(2-g) + \frac{2}{\gamma^2-1} \right] (\vec{\beta} \cdot \vec{s})(\vec{\beta} \cdot \vec{E}) \right. \\ &\quad \left. + [g + \gamma^2(2-g)] (\vec{s} \cdot \vec{E}) \right\} \end{aligned}$$

with $\vec{s} = s_{\parallel}\hat{\beta} + \vec{s}_{\perp}$ and $\vec{E} = E_{\parallel}\hat{\beta} + \vec{E}_{\perp}$,

$$\vec{s} \cdot (\vec{\beta} \times \vec{B}) = \vec{s}_{\perp} \cdot (\vec{\beta} \times \vec{B}), \quad (\vec{\beta} \cdot \vec{s})(\vec{\beta} \cdot \vec{E}) = \beta^2 s_{\parallel} E_{\parallel}, \quad (\vec{s} \cdot \vec{E}) = s_{\parallel} E_{\parallel} + (\vec{s}_{\perp} \cdot \vec{E}_{\perp}),$$

$$\begin{aligned}
\frac{ds_{\parallel}}{dt} &= \frac{q}{2\beta\gamma mc} \left\{ \gamma^2(2-g)[\vec{s}_{\perp} \cdot (\vec{\beta} \times \vec{B})] - \left[\gamma^2(2-g) + \frac{2}{\gamma^2-1} \right] \beta^2 s_{\parallel} E_{\parallel} \right. \\
&\quad \left. + [g + \gamma^2(2-g)] [s_{\parallel} E_{\parallel} + (\vec{s}_{\perp} \cdot \vec{E})] \right\} \\
&= \frac{q}{2\beta\gamma mc} \left\{ \gamma^2(2-g) [\vec{s}_{\perp} \cdot (\vec{\beta} \times \vec{B})] + [g + \gamma^2(2-g)] (\vec{s}_{\perp} \cdot \vec{E}) \right. \\
&\quad \left. + \left[g + \gamma^2(2-g) - (\gamma^2-1)(2-g) - \frac{2}{\gamma^2} \right] s_{\parallel} E_{\parallel} \right\} \\
&= \frac{\gamma q}{mc} \left[\left(\frac{g}{2} - 1 \right) \vec{s}_{\perp} \cdot (\vec{B} \times \hat{\beta}) + \left(\frac{1}{\beta} - \frac{g\beta}{2} \right) (\vec{s}_{\perp} \cdot \vec{E}) + \frac{1}{2\gamma^2} (\vec{\beta} \cdot \vec{E}) s_{\parallel} \right]
\end{aligned}$$

where if $\vec{E} \perp \vec{\beta}$, the last term vanishes.

(b) For the case of $\vec{E} \perp \vec{\beta}$, when

$$\left(\frac{1}{\beta} - \frac{g\beta}{2} \right) = 0 \quad \implies \quad \gamma = \sqrt{\frac{g}{g-2}}$$

the effect of electric field of the polarization of the spin cancels out. For electron or muon, $g \simeq 2.002$, the magic gamma is $\gamma \sim 316$.

HW Assignment 12-1 (12.11, 12.3) and 12-2 (12.5, 12.6)

12.11 Consider the precession of the spin of a muon, initially longitudinally polarized. The muon moves in a circular orbit in a plane perpendicular to a uniform magnetic field.

(a) Precession Frequency and orbital gyration frequency

With only magnetic field, the mechanical energy, and γ and β , of a charged particle are constant. The equation of the orbital motion is

$$m \frac{d\vec{U}}{d\tau} = \frac{e}{c} \vec{U} \times \vec{B} \quad \implies \quad \frac{d\vec{\beta}}{dt} = \frac{e}{\gamma mc} \vec{\beta} \times \vec{B} = -\vec{\omega}_B \times \vec{\beta}$$

where $\vec{\omega}_B = e\vec{B}/(\gamma mc)$ is the cyclotron frequency (orbital gyration frequency) for the orbital motion. Let $\vec{B} = B \vec{e}_z$, the orbital motion is a rotation in the x - y plane,

$$\begin{cases} \beta_x(t) &= \beta_x(0) \cos(\omega_B t) + \beta_y(0) \sin(\omega_B t) \\ \beta_y(t) &= -\beta_y(0) \cos(\omega_B t) + \beta_x(0) \sin(\omega_B t) \\ \beta_z(t) &= 0 \end{cases} \quad \implies \quad \vec{\beta} = -\beta \vec{e}_{\theta}$$

where $\vec{e}_{\theta}$ is unit vector of the counterclockwise angle direction of cylindrical coordinate $(x, y, z) = (r \cos \theta, r \sin \theta, z)$. Note that for a positive charge, $\omega = eB/\gamma mc > 0$ and the orbital rotational motion in $x-y$ plane is clockwise.

Without $\vec{E}$ -field, BMT equation in 3-vector form in the lab frame is

$$\begin{aligned}\frac{d\vec{s}}{dt} &= \frac{e}{\gamma mc} \left\{ \frac{g}{2}(\vec{s} \times \vec{B}) + \frac{(g-2)\gamma^2}{2}[\vec{\beta} \cdot (\vec{s} \times \vec{B})]\vec{\beta} \right\} \\ &= \frac{g}{2}(\vec{s} \times \vec{\omega}_B) + \frac{(g-2)\gamma^2}{2}[\vec{\beta} \cdot (\vec{s} \times \vec{\omega}_B)]\vec{\beta}\end{aligned}$$

where $\vec{B} = B\vec{e}_z$. To simplify the calculation, we write $\vec{s}$ with the cylindrical coordinate that is rotating with the particle. Let

$$\vec{s} = s_r\vec{e}_r + s_\theta\vec{e}_\theta + s_z\vec{e}_z$$

The time derivatives of the base vectors are

$$\frac{d\vec{e}_r}{dt} = \frac{d\theta}{dt} \vec{e}_\theta = -\omega_B \vec{e}_\theta \quad \text{and} \quad \frac{d\vec{e}_\theta}{dt} = -\frac{d\theta}{dt} \vec{e}_r = \omega_B \vec{e}_r,$$

where $d\theta/dt = -\omega_B$ because $\vec{\beta} = -\beta\vec{e}_\theta$. Since

$$\begin{aligned}\frac{d\vec{s}}{dt} &= \frac{ds_z}{dt} \vec{e}_z + \frac{ds_r}{dt} \vec{e}_r + s_r \frac{d\vec{e}_r}{dt} + \frac{ds_\theta}{dt} \vec{e}_\theta + s_\theta \frac{d\vec{e}_\theta}{dt} \\ &= \frac{ds_z}{dt} \vec{e}_z + \frac{ds_r}{dt} \vec{e}_r - s_r\omega_B \vec{e}_\theta + \frac{ds_\theta}{dt} \vec{e}_\theta + s_\theta\omega_B \vec{e}_r\end{aligned}$$

and

$$\begin{cases} \vec{s} \times \vec{\omega}_B &= s_r\omega_B (\vec{e}_r \times \vec{e}_z) + s_\theta\omega_B (\vec{e}_\theta \times \vec{e}_z) = -s_r\omega_B \vec{e}_\theta + s_\theta\omega_B \vec{e}_r \\ [\vec{\beta} \cdot (\vec{s} \times \vec{\omega}_B)] \vec{\beta} &= -s_r\omega_B\beta^2 \vec{e}_\theta \end{cases}$$

BNT equation with the cylindrical coordinate is

$$\begin{cases} \frac{ds_r}{dt} = \left(\frac{g}{2} - 1\right) \omega_B s_\theta \\ \frac{ds_\theta}{dt} = -\left[\left(\frac{g}{2} - 1\right) + \frac{(g-2)\gamma^2\beta^2}{2}\right] \omega_B s_r = -\left(\frac{g}{2} - 1\right) \gamma^2 \omega_B s_r \\ \frac{ds_z}{dt} = 0 \end{cases}$$

The solution of the spin precession for the initial condition of $\vec{s}(0) = s_0 \vec{e}_\theta$ is thus

$$\begin{cases} s_z = 0 \\ s_r = \frac{s_0}{\gamma} \sin(\Omega t) \\ s_\theta = s_0 \cos(\Omega t) \end{cases} \quad \text{and} \quad \Omega = \gamma \omega_B \left(\frac{g}{2} - 1\right) = \frac{eB}{mc} \left(\frac{g}{2} - 1\right)$$

where Ω is the precession frequency which is independent of the energy of muon $E = \gamma mc^2$. This precession frequency can be used to measure g precisely.

(b) For the CERN Muon Storage Ring, the radius is $R = 2.5$ m and $B = 1.7$ T

Integration of $\vec{\beta}(t)$ yields the circular trajectory of muons in the $x - y$ plane as

$$\begin{cases} x(t) &= \frac{1}{\omega_B} [v_x(0) \sin(\omega_B t) - v_y(0) \cos(\omega_B t)] \\ y(t) &= -\frac{1}{\omega_B} [v_y(0) \sin(\omega_B t) + v_x(0) \cos(\omega_B t)] \end{cases}$$

and thus

$$R^2 = x^2(t) + y^2(t) = \frac{(v_x^2 + v_y^2)}{\omega_B^2}$$

The momentum of muon is then

$$p = \gamma m v = \gamma m R \omega_B = \frac{e B}{c} R$$

where the time dilatation factor γ of muon is

$$\gamma = \frac{E}{mc^2} = \frac{1}{mc^2} \sqrt{c^2 p^2 + m^2 c^4} = \sqrt{\left(\frac{p}{mc}\right)^2 + 1}$$

12.3 A particle with mass m and charge e move in a uniform, static field $\vec{E}_0$.

(a) For $\vec{v}(0) \perp \vec{E}_0$, solve the motion.

The equation of motion of the charge in the field is

$$\frac{d\vec{p}}{dt} = e\vec{E}_0 \quad \text{and} \quad \frac{d(\gamma mc^2)}{dt} = e\vec{v} \cdot \vec{E}_0$$

where $\vec{p} = \gamma m \vec{v}$ and γ is not a constant in this case. Since $\vec{E}_0$ is constant,

$$\vec{p}(t) = e\vec{E}_0 t + \vec{p}(0) \quad \text{and} \quad mc^2 \frac{d\gamma}{dt} = \frac{e}{\gamma m} \vec{p} \cdot \vec{E}_0 = \frac{(eE_0)^2}{\gamma m} t$$

where $\vec{p}(0) \perp \vec{E}_0$. From the 2nd equation,

$$\gamma \frac{d\gamma}{dt} = \left(\frac{eE_0}{mc}\right)^2 t \quad \implies \quad \gamma = \left[\left(\frac{eE_0 t}{mc}\right)^2 + \gamma_0^2 \right]^{1/2}$$

To go further, it is convenient to specify a coordinate. Let

$$\vec{E}_0 = E_0 \vec{e}_x \quad \text{and} \quad \vec{v}(0) = v_0 \vec{e}_y$$

Then,

$$\vec{p}(t) = eE_0 t \vec{e}_x + \gamma_0 m v_0 \vec{e}_y \quad \implies \quad \beta_x(t) = \frac{eE_0 t}{\gamma mc}, \quad \beta_y(t) = \frac{\gamma_0 \beta_0}{\gamma}, \quad \beta_z(t) = 0$$

Therefore,

$$\beta_x(t) = \frac{\lambda t}{\sqrt{\lambda^2 t^2 + 1}}, \quad \beta_y(t) = \frac{\beta_0}{\sqrt{\lambda^2 t^2 + 1}}, \quad \beta_z(t) = 0$$

where $\lambda = eE_0/(\gamma_0 mc)$, and

$$\begin{aligned} x(t) &= c \int_0^t \frac{\lambda t}{\sqrt{\lambda^2 t^2 + 1}} dt + x(0) = \frac{c}{\lambda} \left[\sqrt{\lambda^2 t^2 + 1} - 1 \right] + x(0) \\ y(t) &= \int_0^t \frac{v_0}{\sqrt{\lambda^2 t^2 + 1}} dt + y(0) = \frac{v_0}{\lambda} \ln \left(\lambda t + \sqrt{\lambda^2 t^2 + 1} \right) + y(0) \\ z(t) &= z(0) \end{aligned}$$

(b) Obtain the trajectory.

The origin of the coordinate is chosen such that $\vec{r}(0) = 0$. From $x = x(t)$, we have

$$\lambda t = \sqrt{(\lambda x/c + 1)^2 - 1},$$

Substituting this $t = t(x)$ into $y = y(t)$ yields

$$e^{\lambda y/v_0} = \sqrt{(\lambda x/c + 1)^2 - 1} + \lambda x/c + 1 \quad \implies \quad \frac{\lambda x}{c} + 1 = \cosh \left(\frac{\lambda y}{v_0} \right)$$

Short-Time Limit $\lambda t \ll 1$ or $t \ll 1/\lambda$,

$$\begin{aligned} x(t) &= \frac{c}{\lambda} \left[\sqrt{\lambda^2 t^2 + 1} - 1 \right] \simeq \frac{1}{2} c \lambda t^2 \\ y(t) &= \frac{v_0}{\lambda} \ln \left(\lambda t + \sqrt{\lambda^2 t^2 + 1} \right) \simeq v_0 t \end{aligned}$$

where $c\lambda = eE_0/(\gamma_0 m)$ is the classical acceleration. The motion is with a constant acceleration and the trajectory is a parabola

$$x = \frac{c\lambda}{2v_0^2} y^2$$

This is similar to a projectile motion with initial launch along the horizontal (the y axis in our coordinate) direction.

Long-Time Limit $\lambda t \gg 1$ or $t \gg 1/\lambda$,

$$\begin{aligned} x(t) &= \frac{c}{\lambda} \left[\sqrt{\lambda^2 t^2 + 1} - 1 \right] \simeq ct \\ y(t) &= \frac{v_0}{\lambda} \ln(2\lambda t) \simeq v_0 t \end{aligned}$$

In this case, the motion is with a constant velocity and the trajectory is a straight line in the $x - y$ plane.

12.5 A particle moves in the lab frame in crossed, static, uniform electric and magnetic fields, *i.e.* $\vec{E} = E\vec{e}_x$ and $\vec{B} = B\vec{e}_y$.

(a) $E < B$

In K' frame that is moving with a velocity $c\vec{\beta}$ with respect to the lab frame, where

$$\vec{\beta} = \frac{\vec{E} \times \vec{B}}{B^2} = \frac{E}{B} \vec{e}_z \quad \text{and} \quad \gamma = \frac{B}{\sqrt{B^2 - E^2}},$$

the fields are

$$\begin{aligned} \vec{E}' &= \gamma(\vec{E} + \vec{\beta} \times \vec{B}) - \frac{\gamma^2}{\gamma + 1} \vec{\beta}(\vec{\beta} \cdot \vec{E}) = \gamma \left[\vec{E} + \frac{1}{B^2} (\vec{E} \times \vec{B}) \times \vec{B} \right] = 0 \\ \vec{B}' &= \gamma(\vec{B} - \vec{\beta} \times \vec{E}) - \frac{\gamma^2}{\gamma + 1} \vec{\beta}(\vec{\beta} \cdot \vec{B}) = \gamma \left[\vec{B} - \frac{1}{B^2} (\vec{E} \times \vec{B}) \times \vec{E} \right] \\ &= \frac{\sqrt{B^2 - E^2}}{B} \vec{B} = \sqrt{B^2 - E^2} \vec{e}_y \end{aligned}$$

In K' frame, the motion is in a uniform static magnetic field

$$\frac{d\vec{p}'}{dt'} = \frac{e}{c} \vec{v}' \times \vec{B}' \quad \text{and} \quad \frac{d(\gamma_v mc^2)}{dt'} = 0$$

Since $\gamma_v = \gamma_0 = 1/\sqrt{1 - v'^2(0)/c^2}$ is a constant, the equation of motion can be written as

$$\frac{d\vec{v}'}{dt'} = \vec{v}' \times \vec{\omega}_B \quad \text{with} \quad \vec{\omega}_B = \frac{e}{\gamma_v mc} \vec{B}' = \frac{e\sqrt{B^2 - E^2}}{\gamma_0 mc} \vec{e}_y$$

or

$$\dot{v}'_x = -\omega_B v'_z, \quad \dot{v}'_z = \omega_B v'_x, \quad \dot{v}'_y = 0$$

$$\begin{aligned} \vec{v}'(t') &= v'_y(0) \vec{e}_y + [v'_x(0) \cos(\omega_B t') + v'_z(0) \sin(\omega_B t')] \vec{e}_x \\ &\quad + [v'_x(0) \sin(\omega_B t') - v'_z(0) \cos(\omega_B t')] \vec{e}_z \\ &= v'_y(0) \vec{e}_y + \rho \omega_B [\sin(\omega_B t' + \alpha_1) \vec{e}_x - \cos(\omega_B t' + \alpha_1) \vec{e}_z] \end{aligned}$$

where

$$\tan \alpha_1 = \frac{v'_x(0)}{v'_z(0)} \quad \text{and} \quad \rho = \frac{1}{\omega_B} \sqrt{v'^2_x(0) + v'^2_z(0)}$$

and

$$\begin{cases} y'(t') = y_0 + v'_y(0)t' \\ x'(t') = x_0 - \rho \cos(\omega_B t' + \alpha_1) \\ z'(t') = z_0 - \rho \sin(\omega_B t' + \alpha_1) \end{cases}$$

where

$$y_0 = y'(0), \quad x_0 = x'(0) + \frac{v'_z(0)}{\omega_B}, \quad z_0 = z'(0) + \frac{v'_x(0)}{\omega_B}$$

Note that ρ is radius of the gyration in the $x' - z'$ plane. To transfer to the lab frame, we rewrite this solution of trajectory as

$$\begin{cases} y'(t') = y_0 + v'_y(0)t' \\ x'(y') = x_0 - \rho \cos(ky' + \alpha_2) \\ z'(y') = z_0 - \rho \sin(ky' + \alpha_2) \end{cases}$$

where $k = \omega_B/v'_y(0)$ and $\alpha_2 = \alpha_1 - ky_0$. Here we will utilize the fact that y' is the same in the two frames. Since the Lorentz transformation from K' to the lab frame is

$$ct' = \gamma(ct - \beta z), \quad z' = \gamma(z - \beta ct), \quad x' = x, \quad y' = y,$$

the trajectory observed in the lab frame is thus

$$\begin{cases} y = y_0 + v'_y(0)\gamma(t - \beta z/c) \\ x = x_0 - \rho \cos(ky + \alpha_2) \\ z = z'/\gamma + \beta ct = [z_0 - \rho \sin(ky + \alpha_2)]/\gamma + \beta ct \end{cases}$$

where $\beta = E/B$ and $\gamma = B/\sqrt{B^2 - E^2}$. In the case of $\beta z/c \ll 1$,

$$\begin{cases} y = y_0 + \gamma v'_y(0)t \\ x = x_0 - \rho \cos(\gamma \omega_B t + \alpha_1) \\ z = [z_0 - \rho \sin(\gamma \omega_B t + \alpha_1)]/\gamma + \beta ct \end{cases}$$

In the lab frame, therefore, the motion is a combination of circling in the $x - z$ plane and drifting in both the z and y direction.

(b) $E > B$

In K' frame that is moving with a velocity $c\vec{\beta}$ with respect to the lab frame, where

$$\vec{\beta} = \frac{\vec{E} \times \vec{B}}{E^2} = \frac{B}{E} \vec{e}_z \quad \text{and} \quad \gamma = \frac{E}{\sqrt{E^2 - B^2}},$$

the fields are

$$\begin{aligned} \vec{E}' &= \gamma(\vec{E} + \vec{\beta} \times \vec{B}) - \frac{\gamma^2}{\gamma + 1} \vec{\beta}(\vec{\beta} \cdot \vec{E}) = \gamma \left[\vec{E} + \frac{1}{E^2} (\vec{E} \times \vec{B}) \times \vec{B} \right] \\ &= \frac{\sqrt{E^2 - B^2}}{E} \vec{E} = \sqrt{E^2 - B^2} \vec{e}_x \\ \vec{B}' &= \gamma(\vec{B} - \vec{\beta} \times \vec{E}) - \frac{\gamma^2}{\gamma + 1} \vec{\beta}(\vec{\beta} \cdot \vec{B}) = \gamma \left[\vec{B} - \frac{1}{E^2} (\vec{E} \times \vec{B}) \times \vec{E} \right] = 0 \end{aligned}$$

In K' frame, the motion is in a uniform static electric field

$$\frac{d\vec{p}'}{dt'} = e\vec{E}'$$

From the result of Problem 12.3, the trajectory in the K' frame is

$$\frac{\lambda x'}{c} + 1 = \cosh\left(\frac{\lambda y'}{v_0}\right) \quad \text{and} \quad z' = z'(0)$$

where

$$\lambda = \frac{eE'}{\gamma_0 mc} = \frac{e}{\gamma_0 mc} \sqrt{E^2 - B^2} \quad \text{and} \quad \gamma_0 = \frac{1}{\sqrt{1 - v'^2(0)/c^2}}$$

Since the Lorentz transformation from K' to the lab frame is

$$ct' = \gamma(ct - \beta z), \quad z' = \gamma(z - \beta ct), \quad x' = x, \quad y' = y,$$

the trajectory observed in the lab frame is thus

$$\begin{cases} \frac{\lambda x}{c} + 1 = \cosh\left(\frac{\lambda y}{v_0}\right) \\ z(t) = \frac{z'(0)}{\gamma} + \beta ct = z(0) + \frac{cB}{E}t \end{cases}$$

where $z'(0) = \gamma z(0)$.

12.6 Static, uniform electric and magnetic field make an angle of θ with respect to each other.

(b) $\vec{E}$ and $\vec{B}$ are parallel or anti-parallel with each other

We first consider the case of $\theta = 0$, *i.e.* $\vec{E}$ and $\vec{B}$ are parallel or anti-parallel with each other. Let

$$\vec{B} = B \vec{e}_x \quad \text{and} \quad \vec{E} = E \vec{e}_x$$

The equation of motion in the lab frame is then

$$\begin{cases} \frac{dp_x}{dt} = eE \\ \frac{dp_y}{dt} = \frac{e}{c} v_z B \\ \frac{dp_z}{dt} = -\frac{e}{c} v_y B \end{cases} \quad \text{and} \quad \frac{d\gamma}{dt} = \frac{e}{mc^2} v_x E$$

From the equation of dp_x/dt , we have

$$p_x = \gamma m v_x = eEt + p_x(0) \quad \implies \quad v_x = \frac{1}{m\gamma} [eEt + p_x(0)]$$

Substituting this into the equation of $d\gamma/dt$ yields

$$\begin{aligned} \gamma \frac{d\gamma}{dt} &= \frac{eE}{m^2 c^2} [eEt + p_x(0)] \\ \implies \quad \frac{1}{2} [\gamma^2 - \gamma^2(0)] &= \frac{eE}{m^2 c^2} \int_0^t [eEt + p_x(0)] dt = \frac{eE}{2m^2 c^2} [eEt^2 + 2p_x(0)t] \end{aligned}$$

$$\Rightarrow \quad \gamma^2 - \gamma^2(0) = \left(\frac{eE}{mc}\right)^2 \left[t^2 + 2\frac{p_x(0)}{eE}t \right] = \left(\frac{eE}{mc}\right)^2 \left[\left(t + \frac{p_x(0)}{eE}\right)^2 - \left(\frac{p_x(0)}{eE}\right)^2 \right]$$

or

$$\gamma = \frac{eE}{mc} \sqrt{(t+a)^2 + b^2}$$

where

$$a = \frac{p_x(0)}{eE} \quad \text{and} \quad b = \frac{\gamma(0)mc}{eE} \sqrt{1 - v_x^2(0)/c^2}$$

and $\gamma(0)$ is the γ of the initial velocity $\vec{v}(0)$.

For the two equations for dp_y/dt and dp_z/dt , it is convenient to use $d\tau$ instead of dt . With $\gamma d\tau = dt$, they become

$$\frac{dp_y}{d\tau} = \frac{eB}{mc} p_z \quad \text{and} \quad \frac{dp_z}{d\tau} = -\frac{eB}{mc} p_y$$

or

$$\frac{d^2 p_y}{d\tau^2} = -\omega_B^2 p_y \quad \text{where} \quad \omega_B = \frac{eB}{mc}$$

Thus,

$$\begin{aligned} p_y &= p_y(0) \cos(\omega_B \tau) + p_z(0) \sin(\omega_B \tau) \\ p_z &= -p_y(0) \sin(\omega_B \tau) + p_z(0) \cos(\omega_B \tau) \end{aligned}$$

where

$$\begin{aligned} \tau(t) &= \int_0^t \frac{dt}{\gamma} = \frac{mc}{eE} \int_0^t \frac{dt}{\sqrt{(t+a)^2 + b^2}} \\ &= \frac{mc}{eE} \ln \left[\frac{(t+a) + \sqrt{(t+a)^2 + b^2}}{a + \sqrt{a^2 + b^2}} \right] \end{aligned}$$

or

$$\left(a + \sqrt{a^2 + b^2}\right) e^{\rho \omega_B \tau} = (t+a) + \sqrt{(t+a)^2 + b^2}$$

where $\rho = E/B$. To write the trajectory in the form given in the book, let

$$\sinh(\rho\phi) = \frac{t+a}{b} \quad \text{and} \quad \sinh(\rho\phi_0) = \frac{a}{b}$$

Then,

$$\begin{aligned} \left(a + \sqrt{a^2 + b^2}\right) e^{\rho \omega_B \tau} &= b [\sinh(\rho\phi) + \cosh(\rho\phi)] = b e^{\rho\phi} \\ \Rightarrow \quad e^{\rho \omega_B \tau} &= e^{\rho(\phi - \phi_0)} \quad \text{or} \quad \phi = \omega_B \tau + \phi_0 \end{aligned}$$

Thus,

$$t = b \sinh[\rho(\omega_B \tau + \phi_0)] - a$$

To obtain the trajectory, we need the integrations of $\vec{v}$,

$$\begin{aligned}
 y(t) - y(0) &= \int_0^t v_y(t) dt = \frac{1}{m} \int_0^t \frac{p_y[\tau(t)]}{\gamma} dt = \frac{1}{m} \int_0^{\tau(t)} p_y(\tau) d\tau \\
 &= \frac{1}{m\omega_B} [p_y(0) \sin(\omega_B \tau) - p_z(0) \cos(\omega_B \tau)] \\
 &= \frac{1}{m\omega_B} \sqrt{p_y^2(0) + p_z^2(0)} \sin[\omega_B \tau(t) - \alpha_0] \\
 z(t) - z(0) &= \frac{1}{m\omega_B} [p_y(0) \cos(\omega_B \tau) + p_z(0) \sin(\omega_B \tau)] \\
 &= \frac{1}{m\omega_B} \sqrt{p_y^2(0) + p_z^2(0)} \cos[\omega_B \tau(t) - \alpha_0]
 \end{aligned}$$

where $\tan \alpha_0 = p_z(0)/p_y(0)$. Therefore,

$$[y - y(0)]^2 + [z - z(0)]^2 = \frac{1}{m^2 \omega_B^2} [p_y^2(0) + p_z^2(0)]$$

The motion in the $y - z$ plane is a circular motion with a constant radius but a decreasing angular speed. The angular speed of the rotation is

$$\omega = \frac{d(\omega_B \tau)}{dt} = \frac{\omega_B}{\gamma} = \frac{B}{E \sqrt{(t+a)^2 + b^2}}$$

For the trajectory in the x direction,

$$\begin{aligned}
 x(t) - x(0) &= \int_0^t v_x dt = c \int_0^t \frac{(t+a) dt}{\sqrt{(t+a)^2 + b^2}} \\
 &= c \left[\sqrt{(t+a)^2 + b^2} - \sqrt{a^2 + b^2} \right] = cb [\cosh(\rho\phi) - \cosh(\rho\phi_0)]
 \end{aligned}$$

In the case of $v \ll c$ and $\gamma \simeq \gamma(0) \simeq 1$, $t \ll b$ and

$$x(t) - x(0) = \frac{cat}{\sqrt{a^2 + b^2}} + \frac{cb^2 t^2}{2(a^2 + b^2)^{2/3}} + O(t^3) \simeq v_x(0)t + \frac{eE}{2m} t^2$$

To write the trajectory in the form given in the book with $\phi = \omega_B \tau$ as the independent parameter,

$$\begin{cases} t = b \sinh[\rho(\phi + \phi_0)] - a \\ x = x(0) + cb [\cosh(\rho\phi) - \cosh(\rho\phi_0)] \\ y = y(0) + A \sin(\phi - \alpha_0) \\ z = z(0) + A \cos(\phi - \alpha_0) \end{cases} \quad \text{where} \quad A = \frac{1}{m\omega_B} \sqrt{p_y^2(0) + p_z^2(0)}$$

(a) With an Arbitrary θ

We now consider the case that $\vec{E}$ and $\vec{B}$ are making an angle θ with respect to each other. Base on the result of Problem 11.15 (with arbitrary field strengths), $\vec{E}$ and $\vec{B}$ can become parallel or anti-parallel in a moving K' frame. Let

$$\vec{E} = E \vec{e}_x \quad \text{and} \quad \vec{B} = B(\cos \theta \vec{e}_x + \sin \theta \vec{e}_y)$$

and K' frame moves in the z direction with

$$\vec{\beta} = \beta \vec{e}_z \quad \text{if } \theta < \pi/2 \quad \text{and} \quad \vec{\beta} = -\beta \vec{e}_z \quad \text{if } \theta > \pi/2$$

where

$$\beta = \frac{1 - \sqrt{1 - \eta^2}}{\eta} \quad \text{with} \quad \eta = \frac{2EB \sin \theta}{E^2 + B^2} = \frac{2EB_y}{E^2 + B^2} < 1$$

or

$$\vec{\beta} = \frac{1 - \sqrt{1 - \eta^2}}{\eta^2} \vec{\eta} \quad \text{with} \quad \vec{\eta} = \frac{2(\vec{E} \times \vec{B})}{E^2 + B^2}$$

In K' frame, $\vec{E}$ and $\vec{B}$ are parallel or anti-parallel with each other,

$$\begin{aligned} \vec{E}' &= \gamma(\vec{E} + \vec{\beta} \times \vec{B}) = \gamma[(E - \beta B_y)\vec{e}_x + \beta B_x \vec{e}_y] = E' \vec{e}_{x'} \\ \vec{B}' &= \gamma(\vec{B} - \vec{\beta} \times \vec{E}) = \gamma[B_x \vec{e}_x + (B_y - \beta E)\vec{e}_y] = B' \vec{e}_{x'} \end{aligned}$$

where

$$\begin{cases} E' = \gamma[(E - \beta B_y)^2 + \beta^2 B_x^2]^{1/2} \\ B' = \gamma[B_x^2 + (B_y - \beta E)^2]^{1/2} \end{cases} \quad \text{and} \quad \begin{cases} \vec{e}_{x'} = \cos \mu \vec{e}_x + \sin \mu \vec{e}_y \\ \vec{e}_{y'} = -\sin \mu \vec{e}_x + \cos \mu \vec{e}_y \end{cases}$$

with

$$\tan \mu = \frac{E - \beta B_y}{\beta B_x} = \frac{B_x}{B_y - \beta E}$$

Note that the rotation in the $x - y$ plane is to put $\vec{E}'$ and $\vec{B}'$ in the x' direction in K' frame.

In K' frame, the result of part **(b)** can be applied, *i.e.*

$$x' - x'(0) = c \left[\sqrt{(t' + a)^2 + b^2} - \sqrt{a^2 + b^2} \right]$$

$$[y' - y'(0)]^2 + [z' - z'(0)]^2 = \frac{1}{m^2 \omega_B^2} [p_y'^2(0) + p_z'^2(0)]$$

where all the constants are calculated with quantities in K' frame. The Lorentz transformation from K' to the lab frame without the rotation of the coordinate in the $x - y$ plane with an angle μ is

$$ct' = \gamma(ct - \beta z), \quad z' = \gamma(z - \beta ct), \quad x' = x, \quad y' = y,$$

Thus, without the rotation of x and y back to their original orientation, the trajectory in the Lab frame is

$$\begin{cases} x = x'(0) + c \left[\sqrt{(\gamma t + \gamma \beta z/c + a)^2 + b^2} - \sqrt{a^2 + b^2} \right] \\ [y - y'(0)]^2 + [\gamma z - \gamma \beta c t - z'(0)]^2 = \frac{1}{m^2 \omega_B^2} [p_y'^2(0) + p_z'^2(0)] \end{cases}$$

where (x, y, z) are of $\vec{r} = x\vec{e}_{x'} + y\vec{e}_{y'} + z\vec{e}_{z'}$ in the lab frame.

HW Assignment 14-1 (14.1, 14.3, 14.4)

14.1

Consider a charge moving with a constant velocity in the x direction, for $\dot{\vec{\beta}} = 0$, the field emitted from the charge is

$$\vec{E}(\vec{x}, t) = q \left[\frac{\vec{n} - \vec{\beta}}{\gamma^2 R^2 (1 - \vec{n} \cdot \vec{\beta})^3} \right]_{ret} \quad \text{and} \quad \vec{B}(\vec{x}, t) = (\vec{n} \times \vec{E})_{ret}$$

Using the coordinate in Fig. 14.2 in Jackson's book,

$$\vec{n} = \cos \theta \vec{e}_x + \sin \theta \vec{e}_y \quad \text{and} \quad \vec{\beta} = \beta \vec{e}_x$$

$$1 - \vec{n} \cdot \vec{\beta} = 1 - \beta \cos \theta \quad \text{and} \quad \vec{n} - \vec{\beta} = (\cos \theta - \beta) \vec{e}_x + \sin \theta \vec{e}_y$$

where

$$\begin{aligned} \cos \theta &= \frac{\beta R + vt}{R} = \beta + \frac{vt}{R} \quad \text{and} \quad \sin \theta = \frac{b}{R} \\ R^2 &= b^2 + (\beta R + vt)^2 \quad \implies \quad R = \gamma^2 \beta vt + \gamma \sqrt{b^2 + \gamma^2 v^2 t^2} \\ \implies \quad 1 - \vec{n} \cdot \vec{\beta} &= \frac{1}{\gamma^2} - \beta \frac{vt}{R} = \frac{1}{\gamma R} \sqrt{b^2 + \gamma^2 v^2 t^2} \end{aligned}$$

$$E_x = q \frac{R\gamma(\cos \theta - \beta)}{(b^2 + \gamma^2 v^2 t^2)^{3/2}} = \frac{\gamma q v t}{(b^2 + \gamma^2 v^2 t^2)^{3/2}}$$

$$E_y = q \frac{R\gamma \sin \theta}{(b^2 + \gamma^2 v^2 t^2)^{3/2}} = \frac{\gamma q b}{(b^2 + \gamma^2 v^2 t^2)^{3/2}}$$

$$E_z = 0$$

$$B_x = 0$$

$$B_y = 0$$

$$B_z = n_x E_y - n_y E_x = q \frac{R\gamma \beta \sin \theta}{(b^2 + \gamma^2 v^2 t^2)^{3/2}} = \frac{q\gamma \beta b}{(b^2 + \gamma^2 v^2 t^2)^{3/2}} = \beta E_y$$

Which is the same field calculated from a charge moving with a constant velocity in the x direction [Eq. (11.152) in Chapter 11 of Jackson's book].

14.3 Derive the Lienard-Wiechert field from the Heaviside-Feynman expression,

$$\vec{E} = e \left(\frac{\vec{n}}{R^2} \right)_{ret} + e \left(\frac{R}{c} \right)_{ret} \frac{d}{dt} \left(\frac{\vec{n}}{R^2} \right)_{ret} + e \frac{d^2}{c^2 dt^2} [\vec{n}]_{ret}$$

where the retardation condition is

$$t' = t - \frac{1}{c} R(t'), \quad R(t') = |\vec{x} - \vec{r}(t')|, \quad \text{and} \quad \vec{n} = \frac{1}{R} [\vec{x} - \vec{r}(t')]$$

Since

$$\begin{aligned} \frac{dR}{dt'} &= -c \vec{\beta} \cdot \vec{n} \\ \frac{dt}{dt'} &= 1 + \frac{1}{c} \frac{dR}{dt'} = 1 - \vec{\beta} \cdot \vec{n} \\ \frac{d\vec{n}}{dt'} &= c \frac{(\vec{\beta} \cdot \vec{n})\vec{n} - \vec{\beta}}{R} = \frac{c}{R} [\vec{n} \times (\vec{n} \times \vec{\beta})] \\ \frac{d\vec{n}}{dt} &= \frac{1}{(1 - \vec{\beta} \cdot \vec{n})} \frac{d\vec{n}}{dt'} = c \frac{\vec{n} \times (\vec{n} \times \vec{\beta})}{R(1 - \vec{\beta} \cdot \vec{n})} \end{aligned}$$

$$\frac{d}{dt} \left(\frac{\vec{n}}{R^2} \right)_{ret} = \frac{1}{(1 - \vec{\beta} \cdot \vec{n})} \frac{d}{dt'} \left(\frac{\vec{n}}{R^2} \right)_{ret} = \frac{c}{R^3(1 - \vec{\beta} \cdot \vec{n})} [3(\vec{\beta} \cdot \vec{n})\vec{n} - \vec{\beta}]$$

and

$$\frac{d^2 \vec{n}}{dt^2} = \frac{c}{(1 - \vec{\beta} \cdot \vec{n})} \frac{d}{dt'} \left[\frac{\vec{n} \times (\vec{n} \times \vec{\beta})}{R(1 - \vec{\beta} \cdot \vec{n})} \right]$$

where

$$\begin{aligned} \frac{d}{dt'} [\vec{n} \times (\vec{n} \times \vec{\beta})] &= \frac{d\vec{n}}{dt'} \times (\vec{n} \times \vec{\beta}) + \vec{n} \times \left(\frac{d\vec{n}}{dt'} \times \vec{\beta} \right) + \vec{n} \times (\vec{n} \times \dot{\vec{\beta}}) \\ &= \frac{c}{R} \left\{ [(\vec{\beta} \cdot \vec{n})\vec{n} - \vec{\beta}] \times (\vec{n} \times \vec{\beta}) + \vec{n} \times \left([(\vec{\beta} \cdot \vec{n})\vec{n} - \vec{\beta}] \times \vec{\beta} \right) \right\} + \vec{n} \times (\vec{n} \times \dot{\vec{\beta}}) \\ &= \frac{c}{R} \left\{ 2(\vec{\beta} \cdot \vec{n}) [\vec{n} \times (\vec{n} \times \vec{\beta})] - \vec{\beta} \times (\vec{n} \times \vec{\beta}) \right\} + \vec{n} \times (\vec{n} \times \dot{\vec{\beta}}) \\ &= \frac{c}{R} \left\{ [2(\vec{\beta} \cdot \vec{n})^2 - \beta^2] \vec{n} - (\vec{\beta} \cdot \vec{n})\vec{\beta} \right\} + \vec{n} \times (\vec{n} \times \dot{\vec{\beta}}) \\ \frac{d}{dt'} \frac{1}{R(1 - \vec{\beta} \cdot \vec{n})} &= -\frac{1}{R^2(1 - \vec{\beta} \cdot \vec{n})^2} \left[(1 - \vec{\beta} \cdot \vec{n}) \frac{dR}{dt'} - R \left(\dot{\vec{\beta}} \cdot \vec{n} + \vec{\beta} \cdot \frac{d\vec{n}}{dt'} \right) \right] \\ &= \frac{1}{R^2(1 - \vec{\beta} \cdot \vec{n})^2} \left\{ c(1 - \vec{\beta} \cdot \vec{n})(\vec{\beta} \cdot \vec{n}) + R\dot{\vec{\beta}} \cdot \vec{n} + c[(\vec{\beta} \cdot \vec{n})^2 - \beta^2] \right\} \\ &= \frac{c}{R^2(1 - \vec{\beta} \cdot \vec{n})^2} \left(\vec{\beta} \cdot \vec{n} - \beta^2 + \frac{R}{c} \dot{\vec{\beta}} \cdot \vec{n} \right) \end{aligned}$$

$$\begin{aligned}
\frac{d}{dt'} \left[\frac{\vec{n} \times (\vec{n} \times \vec{\beta})}{R(1 - \vec{\beta} \cdot \vec{n})} \right] &= \frac{1}{R(1 - \vec{\beta} \cdot \vec{n})} \frac{d}{dt'} \left[\vec{n} \times (\vec{n} \times \vec{\beta}) \right] + \left[\vec{n} \times (\vec{n} \times \vec{\beta}) \right] \frac{d}{dt'} \frac{1}{R(1 - \vec{\beta} \cdot \vec{n})} \\
&= \frac{c}{R^2(1 - \vec{\beta} \cdot \vec{n})} \left\{ 2(\vec{\beta} \cdot \vec{n}) \left[\vec{n} \times (\vec{n} \times \vec{\beta}) \right] - \vec{\beta} \times (\vec{n} \times \vec{\beta}) \right\} \\
&\quad + \frac{c}{R^2(1 - \vec{\beta} \cdot \vec{n})^2} \left[\vec{n} \times (\vec{n} \times \vec{\beta}) \right] \left(\vec{\beta} \cdot \vec{n} - \beta^2 + \frac{R}{c} \dot{\vec{\beta}} \cdot \vec{n} \right) + \frac{\vec{n} \times (\vec{n} \times \dot{\vec{\beta}})}{R(1 - \vec{\beta} \cdot \vec{n})^2}
\end{aligned}$$

Thus

$$\begin{aligned}
\frac{1}{c^2} \frac{d^2 \vec{n}}{dt'^2} &= \frac{1}{R^2(1 - \vec{\beta} \cdot \vec{n})^2} \left\{ 2(\vec{\beta} \cdot \vec{n}) \left[\vec{n} \times (\vec{n} \times \vec{\beta}) \right] - \vec{\beta} \times (\vec{n} \times \vec{\beta}) \right\} \\
&\quad + \frac{1}{R^2(1 - \vec{\beta} \cdot \vec{n})^3} \left[\vec{n} \times (\vec{n} \times \vec{\beta}) \right] \left(\vec{\beta} \cdot \vec{n} - \beta^2 + \frac{R}{c} \dot{\vec{\beta}} \cdot \vec{n} \right) + \frac{\vec{n} \times (\vec{n} \times \dot{\vec{\beta}})}{cR(1 - \vec{\beta} \cdot \vec{n})^3}
\end{aligned}$$

and

$$\begin{aligned}
\left(\frac{R}{c} \right)_{ret} \frac{d}{dt} \left(\frac{\vec{n}}{R^2} \right)_{ret} + \frac{d^2 [\vec{n}]_{ret}}{c^2 dt^2} &= \frac{1}{R^2(1 - \vec{\beta} \cdot \vec{n})} \left[\vec{n} \times (\vec{n} \times \vec{\beta}) + 2(\vec{\beta} \cdot \vec{n}) \vec{n} \right] \\
&\quad + \frac{1}{R^2(1 - \vec{\beta} \cdot \vec{n})^2} \left\{ 2(\vec{\beta} \cdot \vec{n}) \left[\vec{n} \times (\vec{n} \times \vec{\beta}) \right] - \vec{\beta} \times (\vec{n} \times \vec{\beta}) \right\} \\
&\quad + \frac{1}{R^2(1 - \vec{\beta} \cdot \vec{n})^3} \left[\vec{n} \times (\vec{n} \times \vec{\beta}) \right] \left(\vec{\beta} \cdot \vec{n} - \beta^2 + \frac{R}{c} \dot{\vec{\beta}} \cdot \vec{n} \right) + \frac{\vec{n} \times (\vec{n} \times \dot{\vec{\beta}})}{cR(1 - \vec{\beta} \cdot \vec{n})^3} \\
&= -\frac{\vec{n}}{R^2} + \frac{1}{R^2(1 - \vec{\beta} \cdot \vec{n})^3} \left[\vec{n} \times (\vec{n} \times \vec{\beta}) \right] \left(1 - \beta^2 + \frac{R}{c} \dot{\vec{\beta}} \cdot \vec{n} \right) + \frac{\vec{n} \times (\vec{n} \times \dot{\vec{\beta}})}{cR(1 - \vec{\beta} \cdot \vec{n})^3}
\end{aligned}$$

Substituting this to the Heaviside-Feynman expression of $\vec{E}$ yields

$$\begin{aligned}
\vec{E} &= \frac{q}{R^2(1 - \vec{\beta} \cdot \vec{n})^3} \left[\vec{n} \times (\vec{n} \times \vec{\beta}) \right] \left(1 - \beta^2 + \frac{R}{c} \dot{\vec{\beta}} \cdot \vec{n} \right) + q \frac{\vec{n} \times (\vec{n} \times \dot{\vec{\beta}})}{cR(1 - \vec{\beta} \cdot \vec{n})^3} \\
&= \frac{q}{R^2(1 - \vec{\beta} \cdot \vec{n})^3} \left[(\vec{n} \cdot \vec{\beta}) \vec{n} - \vec{\beta} \right] \left(\frac{1}{\gamma^2} + \frac{R}{c} \dot{\vec{\beta}} \cdot \vec{n} \right) + q \frac{\vec{n} \times (\vec{n} \times \dot{\vec{\beta}})}{cR(1 - \vec{\beta} \cdot \vec{n})^3}
\end{aligned}$$

14.4

Calculate the time-averaged power radiation per unit solid angle in non-relativistic motion of a particle with charge e .

a) Particle moving along the z axis with

$$z(t) = a \cos \omega_0 t$$

For $\beta \ll 1$, the power radiation per unit solid angle is

$$\frac{dP(t')}{d\Omega} = \frac{e^2}{4\pi c} \left| \vec{n} \times (\vec{n} \times \dot{\vec{\beta}}) \right|^2$$

where

$$\dot{\vec{\beta}} = \frac{1}{c} \frac{d^2 z(t')}{dt'^2} \vec{e}_z = -\frac{1}{c} (a\omega_0^2 \cos \omega_0 t) \vec{e}_z$$

Let

$$\begin{aligned} \vec{n} &= (\cos \phi \sin \theta) \vec{e}_x + (\sin \phi \sin \theta) \vec{e}_y + (\cos \theta) \vec{e}_z \\ \vec{n} \times (\vec{n} \times \dot{\vec{\beta}}) &= (\vec{n} \cdot \dot{\vec{\beta}}) \vec{n} - \dot{\vec{\beta}} \end{aligned}$$

and

$$\left| \vec{n} \times (\vec{n} \times \dot{\vec{\beta}}) \right|^2 = (\vec{n} \cdot \dot{\vec{\beta}})^2 + \dot{\beta}^2 - 2(\vec{n} \cdot \dot{\vec{\beta}}) \vec{n} \cdot \dot{\vec{\beta}} = \dot{\beta}^2 - (\vec{n} \cdot \dot{\vec{\beta}})^2 = \dot{\beta}^2 \sin^2 \theta$$

Thus

$$\frac{dP(t')}{d\Omega} = \frac{e^2 a^2 \omega_0^4}{4\pi c^3} \sin^2 \theta \cos^2 \omega_0 t'$$

and the time average power is

$$\left\langle \frac{dP(t')}{d\Omega} \right\rangle_{t'} = \frac{e^2 a^2 \omega_0^3}{8\pi c^3} \sin^2 \theta$$

b) Particle moving in a circle of radius R in the $x - y$ plane.

$$\dot{\beta} = \frac{R\omega^2}{c} \quad \text{and} \quad \dot{\vec{\beta}} = -\dot{\beta}(\vec{e}_x \cos \omega t + \vec{e}_y \sin \omega t)$$

where ω is the angular speed of the particle.

$$\begin{aligned} \left| \vec{n} \times (\vec{n} \times \dot{\vec{\beta}}) \right|^2 &= \dot{\beta}^2 - (\vec{n} \cdot \dot{\vec{\beta}})^2 = \dot{\beta}^2 [1 - (\cos \phi \sin \theta \cos \omega t + \sin \phi \sin \theta \sin \omega t)^2] \\ &= \dot{\beta}^2 [1 - \cos^2(\phi - \omega t) \sin^2 \theta] \\ \left\langle \frac{dP(t')}{d\Omega} \right\rangle_{t'} &= \frac{e^2 R^2 \omega^3}{4\pi c^3} \left(1 - \frac{1}{2} \sin^2 \theta \right) \end{aligned}$$

HW Assignment 14-2 (14.5, 14.7, 14.9)

14.5 A nonrelativistic charged particle makes a head-on collision with a fixed central force field of finite range.

a) For a head-on collision with zero impact parameter, the angular momentum is zero and the Hamiltonian of the particle motion is

$$H = \frac{1}{2m} p_r^2 + V(r)$$

The equation of motion is

$$m\ddot{r} = -\frac{\partial V}{\partial r}$$

In the case of $\beta \ll 1$, the radiation power during the collision is

$$\frac{dP(t')}{d\Omega} = \frac{q^2}{4\pi c} \left| \vec{n} \times (\vec{n} \times \dot{\vec{\beta}}) \right|^2$$

Choose a coordinate such that the particle travels along and collides head-on in the z direction, thus

$$\dot{\vec{\beta}} = - \left(\frac{1}{mc} \frac{\partial V}{\partial r} \right) \vec{e}_z$$

Let

$$\vec{n} = (\cos \phi \sin \theta) \vec{e}_x + (\sin \phi \sin \theta) \vec{e}_y + (\cos \theta) \vec{e}_z$$

Then

$$\left| \vec{n} \times (\vec{n} \times \dot{\vec{\beta}}) \right|^2 = \dot{\beta}^2 - (\vec{n} \cdot \dot{\vec{\beta}})^2 = \dot{\beta}^2 \sin^2 \theta = \left(\frac{1}{mc} \frac{\partial V}{\partial r} \right)^2 \sin^2 \theta$$

and

$$\frac{dP(t')}{d\Omega} = \frac{q^2}{4\pi m^2 c^3} \left(\frac{\partial V}{\partial r} \right)^2 \sin^2 \theta$$

The total radiation power is

$$P(t') = \frac{q^2}{4\pi m^2 c^3} \left(\frac{\partial V}{\partial r} \right)^2 \int \sin^2 \theta d\Omega = \frac{2q^2}{3m^2 c^3} \left(\frac{\partial V}{\partial r} \right)^2$$

The total energy radiated is then

$$\begin{aligned} \Delta W &= \int_{-\infty}^{\infty} P(t') dt' = \frac{2q^2}{3m^2 c^3} \int_{-\infty}^{\infty} \left\{ \frac{\partial V[r(t')]}{\partial r(t')} \right\}^2 dt' \\ &= 2 \frac{2q^2}{3m^2 c^3} \int_{r_{min}}^{\infty} \left[\frac{\partial V(r)}{\partial r} \right]^2 \left(\frac{dt}{dr} \right) dr \end{aligned}$$

where the motion from $r = \infty$ at $t = -\infty$ to $r = r_{min}$ at $t = 0$ is symmetrical to the motion from $r = r_{min}$ at $t = 0$ back to $r = \infty$ at $t = \infty$. To obtaining dt/dr , we go back to the equation of motion. Since

$$m\ddot{r} = \frac{m}{2} \frac{d\dot{r}^2}{dr} = -\frac{\partial V}{\partial r} \quad \implies \quad \dot{r}^2(r) - \dot{r}^2(r_{min}) = -\frac{2}{m} [V(r) - V(r_{min})]$$

and at $r = r_{min}$, $\dot{r}(r_{min}) = 0$,

$$\frac{dt}{dr} = \sqrt{\frac{m}{2}} \frac{1}{\sqrt{V(r_{min}) - V(r)}}$$

and

$$\Delta W = \frac{4q^2}{3m^2 c^3} \sqrt{\frac{m}{2}} \int_{r_{min}}^{\infty} \left[\frac{\partial V(r)}{\partial r} \right]^2 \frac{dr}{\sqrt{V(r_{min}) - V(r)}}$$

b) For a Coulomb potential

$$V(r) = \frac{zZe^2}{r}$$

r_{min} can be solved from

$$\frac{1}{2}mv_0^2 = V(r_{min}) = E \quad \Rightarrow \quad r_{min} = \frac{zZe^2}{E}$$

where the left side equation is the total energy at $r = \infty$ and the middle is the total energy at $r = r_{min}$ and E is the total energy.

$$\Delta W = \frac{4z^2e^2}{3m^2c^3} \sqrt{\frac{m}{2E}} (zZe^2)^2 \int_{r_{min}}^{\infty} \frac{1}{r^4} \frac{dr}{\sqrt{1 - zZe^2/(Er)}}$$

Let

$$x = \frac{zZe^2}{Er} \quad \text{and} \quad dx = -\frac{zZe^2}{Er^2} dr$$

Then

$$\begin{aligned} \Delta W &= \frac{4z^2e^2}{3m^2c^3} \sqrt{\frac{m}{2E}} (zZe^2)^2 \left(\frac{E}{zZe^2} \right)^3 \int_0^{x_{max}} \frac{x^2 dx}{\sqrt{1-x}} \\ &= \frac{zm v_0^5}{6Zc^3} \int_0^{x_{max}} \frac{x^2 dx}{\sqrt{1-x}} = \frac{8zm v_0^5}{45Zc^3} \end{aligned}$$

14.7 A nonrelativistic particle of charge ze , mass m and initial speed v_0 is incident on a fixed charge Ze at an impact parameter b that is large enough to ensure a small angle deflection.

(a) Using the Larmor power formula to calculate the total energy radiated.

The radiation power is

$$P = \frac{2z^2e^2}{3m^2c^3} \left(\frac{d\vec{p}}{dt} \right)^2$$

where (see Lecture note Chapter 15. Bremsstrahlung)

$$\frac{d\vec{p}}{dt} = \frac{zZe^2}{r^3} \vec{r} = \frac{zZe^2}{[(v_0t)^2 + b^2]^{3/2}} (\vec{v}_0t + \vec{b})$$

and

$$\left| \frac{d\vec{p}}{dt} \right|^2 = \frac{z^2Z^2e^4}{[(v_0t)^2 + b^2]^2}$$

The total energy radiated is thus

$$\begin{aligned} \Delta W = \int_{-\infty}^{\infty} P dt &= \frac{2z^4Z^2e^6}{3m^2c^3b^4} \int_{-\infty}^{\infty} \frac{1}{[(v_0t/b)^2 + 1]^2} dt \\ &= \frac{2z^4Z^2e^6}{3m^2c^3b^3v_0} \int_{-\infty}^{\infty} \frac{1}{(x^2 + 1)^2} dx = \frac{\pi z^4Z^2e^6}{3m^2c^3b^3v_0} \end{aligned}$$

(b) For a head-on collision with $b = 0$ and a repulsive interaction potential, the closet distance between the incident particle and target center is r_c where

$$\frac{1}{2}mv_0^2 = \frac{zZe^2}{r_c} \quad \Longrightarrow \quad r_c = \frac{2zZe^2}{mv_0^2}$$

The energy radiated for the head-on collision is calculated in Problem 14-5 as

$$\Delta W_{\text{headon}} = \frac{8zm v_0^5}{45Zc^3}$$

The part (a) result is valid only for $b \gg r_c$. If substituting $b = r_c$ in ΔW calculated in Part (a),

$$\begin{aligned} \Delta W_{b=r_c} &= \frac{\pi z^4 Z^2 e^6}{3m^2 c^3 v_0} \left(\frac{mv_0^2}{2zZe^2} \right)^3 = \frac{\pi z m v_0^5}{24Zc^3} \\ \frac{W_{b=r_c}}{W_{\text{headon}}} &= \frac{45\pi}{8 \times 24} \simeq 0.74 \end{aligned}$$

(c) Radiation Cross Section χ

$$\chi = \int_{b_{\min}}^{\infty} \Delta W (2\pi b) db = \frac{2\pi^2 z^4 Z^2 e^6}{3m^2 c^3 v_0} \int_{b_{\min}}^{\infty} \frac{db}{b^2} = \frac{2\pi^2 z^4 Z^2 e^6}{3m^2 c^3 v_0 b_{\min}}$$

where $b_{\min}$ is the cutoff for the impact parameter. From the quantum mechanics consideration, the smallest b corresponds to the minimal angular momentum of $p_\theta = \hbar$. Since the angular momentum of central force problem is

$$p_\theta = bp = bmv_0 \quad \Longrightarrow \quad b_{\min} = \frac{\hbar}{mv_0} \quad \Longrightarrow \quad \chi = \frac{2\pi^2 z^4 Z^2 e^6}{3mc^3 \hbar}$$

which is the similar to the result of the nonrelativistic Bremsstrahlung.

14.9 A particle of mass m , charge q , moves in a plane perpendicular to a uniform, static magnetic field B . The equation of motion is

$$\frac{d(\gamma m \vec{v})}{dt} = \frac{q}{c} \vec{v} \times \vec{B} \quad \text{and} \quad \frac{d(\gamma mc^2)}{dt} = e\vec{v} \cdot \vec{E} = 0$$

Therefore, the particle moves in a circular orbit with the acceleration and frequency,

$$\frac{dv}{dt} = \frac{qvB}{\gamma mc} \quad \text{and} \quad \omega = \frac{qB}{\gamma mc}$$

(a) Total Radiation Energy

From the Lecture Note, the radiation power per solid angle in the case of circular motion is

$$\frac{dP(t')}{d\Omega} = \frac{q^2}{4\pi c^3} \frac{\dot{v}^2}{(1 - \beta \cos \theta)^3} \left[1 - \frac{\sin^2 \theta \sin^2 \phi}{\gamma^2 (1 - \beta \cos \theta)^2} \right]$$

The total radiation power (energy per unit time) is

$$\begin{aligned}
 P(t') &= \int \frac{dP(t')}{d\Omega} d\Omega \\
 &= \frac{q^2}{4\pi c^3} \dot{v}^2 \int_0^\pi \int_0^{2\pi} \left[\frac{1}{(1 - \beta \cos \theta)^3} - \frac{\sin^2 \theta \sin^2 \phi}{\gamma^2 (1 - \beta \cos \theta)^5} \right] \sin \theta d\theta d\phi \\
 &= \frac{q^2}{4\pi c^3} \dot{v}^2 \left(4\pi \gamma^4 - \frac{4\pi}{3} \gamma^4 \right) = \frac{2q^2 \gamma^4}{3c^3} \dot{v}^2 \\
 &= \frac{2q^4 B^2 \gamma^2 \beta^2}{3m^2 c^3} = \frac{2q^4 B^2}{3m^2 c^3} (\gamma^2 - 1)
 \end{aligned}$$

(b) As the particle has a total energy of $E = \gamma mc^2$ at time t , the rate of the energy loss due to the radiation is

$$\frac{dE}{dt} = mc^2 \frac{d\gamma}{dt} = -P$$

Thus

$$t = -mc^2 \int_{\gamma_0}^{\gamma} \frac{d\gamma}{P} = -mc^2 \frac{3m^2 c^3}{2q^4 B^2} \int_{\gamma_0}^{\gamma} \frac{d\gamma}{(\gamma^2 - 1)}$$

where $E_0 = \gamma_0 mc^2$ is the initial energy at $t = 0$. For $\gamma \gg 1$,

$$t = -\frac{3m^3 c^5}{2q^4 B^2} \int_{\gamma_0}^{\gamma} \frac{d\gamma}{\gamma^2} = \frac{3m^3 c^5}{2q^4 B^2} \left(\frac{1}{\gamma} - \frac{1}{\gamma_0} \right)$$

(c) If the particle is nonrelativistic and has a kinetic energy T_0 at $t = 0$.

For $\beta \ll 1$,

$$P = \frac{2q^4 B^2}{3m^2 c^3} (\gamma^2 - 1) \simeq \frac{2q^4 B^2 \beta^2}{3m^2 c^3} = \frac{4q^4 B^2}{3m^3 c^5} T$$

where $T = mv^2/2$ is the kinetic energy. On the other hand, at $\beta \ll 1$,

$$E = \gamma mc^2 = \frac{mc^2}{\sqrt{1 - \beta^2}} \simeq mc^2 \left(1 + \frac{1}{2} \beta^2 \right) = mc^2 + T$$

and

$$\frac{dE}{dt} = \frac{dT}{dt} = -P = -\frac{4q^4 B^2}{3m^3 c^5} T$$

Thus

$$\int_{T_0}^T \frac{dT}{T} = -\frac{4q^4 B^2}{3m^3 c^5} t \quad \Rightarrow \quad T = T_0 \exp \left(-\frac{4q^4 B^2}{3m^3 c^5} t \right)$$

(d) If a charged particle is trapped in the magnetic dipole field of the earth and is spiraling back and forth along a magnetic field line. It radiates more while near its turning points near the poles than near the equator because B is larger near the poles.

HW Assignment 14-3 (14.10, 14.11, 14.12)

14.10

During a short time interval $0 < t < \Delta t$, the speed of a particle of charge e decreases linearly in time to zero.

(a) The Radiation Energy Per Solid Angle

The radiation power per solid angle is

$$\frac{dP(t')}{d\Omega} = \frac{e^2}{4\pi c} \frac{|\vec{n} \times [(\vec{n} - \vec{\beta}) \times \dot{\vec{\beta}}]|^2}{(1 - \vec{\beta} \cdot \vec{n})^5}$$

and the t' -dependences of acceleration and velocity are

$$\dot{\vec{\beta}}(t') = \begin{cases} -\vec{\beta}_0/\Delta t & \text{for } 0 < t' < \Delta t \\ 0 & \text{otherwise} \end{cases}$$

and for $0 < t' < \Delta t$

$$\vec{\beta}(t') = \vec{\beta}_0 \left(1 - \frac{t'}{\Delta t}\right)$$

where $\vec{\beta}_0$ is the initial velocity of the particle. Since $\vec{\beta} \parallel \dot{\vec{\beta}}$,

$$|\vec{n} \times [(\vec{n} - \vec{\beta}) \times \dot{\vec{\beta}}]|^2 = |(\dot{\vec{\beta}} \cdot \vec{n})\vec{n} - \dot{\vec{\beta}}|^2 = \dot{\beta}^2 - (\dot{\vec{\beta}} \cdot \vec{n})^2$$

Let $\vec{\beta}_0 \cdot \vec{n} = \beta_0 \cos \theta$. For $0 < t' < \Delta t$,

$$\frac{dP(t')}{d\Omega} = \frac{e^2}{4\pi c} \frac{\dot{\beta}^2(1 - \cos^2 \theta)}{(1 - \beta \cos \theta)^5} = \frac{e^2 \beta_0^2}{4\pi c (\Delta t)^2} \frac{\sin^2 \theta}{[1 - \beta(t') \cos \theta]^5}$$

For $t' < 0$ or $t' \geq \Delta t$, $dP(t')/d\Omega = 0$. The radiation energy per solid angle is thus

$$\begin{aligned} \frac{dE}{d\Omega} &= \int_{-\infty}^{\infty} \frac{dP(t')}{d\Omega} dt' = \frac{e^2 \beta_0^2 \sin^2 \theta}{4\pi c (\Delta t)^2} \int_0^{\Delta t} \frac{1}{(1 - \beta(t') \cos \theta)^5} dt' \\ &= \frac{e^2 \beta_0^2 \sin^2 \theta}{4\pi c (\Delta t)^2} \int_{\beta_0}^0 \frac{1}{(1 - \beta \cos \theta)^5} \left(\frac{d\beta}{dt}\right)^{-1} d\beta \\ &= \frac{e^2 \beta_0 \sin^2 \theta}{4\pi c \Delta t} \int_0^{\beta_0} \frac{d\beta}{(1 - \beta \cos \theta)^5} = \frac{e^2 \beta_0 \sin^2 \theta}{16\pi c \Delta t \cos \theta} \left[\frac{1}{(1 - \beta_0 \cos \theta)^4} - 1 \right] \\ &\stackrel{\beta_0 \rightarrow \beta}{=} \frac{e^2 \beta \sin^2 \theta}{16\pi c \Delta t \cos \theta} \left[\frac{1}{(1 - \beta \cos \theta)^4} - 1 \right] \end{aligned}$$

(b) For $\gamma \gg 1$,

$$\beta = \sqrt{1 - \frac{1}{\gamma^2}} \simeq 1 - \frac{1}{2\gamma^2}$$

and the radiation is concentrated in a small cone of $\theta \sim 0$, i.e.

$$\cos \theta \simeq 1 - \frac{1}{2}\theta^2$$

and

$$1 - \beta \cos \theta \simeq 1 - \left(1 - \frac{1}{2\gamma^2}\right) \left(1 - \frac{1}{2}\theta^2\right) \simeq \frac{1}{2\gamma^2} + \frac{1}{2}\theta^2 = \frac{1 + \gamma^2\theta^2}{2\gamma^2} \ll 1$$

$$\frac{dE}{d\Omega} \simeq \frac{e^2\beta\gamma^6}{\pi c\Delta t} \frac{(\gamma\theta)^2}{(1 + \gamma^2\theta^2)^4}$$

$$dE \simeq \frac{e^2\beta\gamma^6}{\pi c\Delta t} \frac{(\gamma\theta)^2}{(1 + \gamma^2\theta^2)^4} \sin \theta d\theta \int_0^{2\pi} d\phi \simeq \frac{e^2\beta\gamma^4}{c\Delta t} \frac{(\gamma\theta)^2}{(1 + \gamma^2\theta^2)^4} d(\gamma\theta)^2$$

Thus

$$\frac{dE}{d\xi} = \frac{e^2\beta\gamma^4}{c\Delta t} \frac{\xi}{(1 + \xi)^4}$$

14.11 A particle of charge $q = ze$ and mass m moves in external electric $\vec{E}$ and magnetic field $\vec{B}$. Derive the total radiation power?

(a) Radiation energy per unit solid angle from a moving charge q can be calculated based on the retarded time as

$$\frac{dP(t_r)}{d\Omega} = \frac{q^2}{4\pi c} \frac{|\vec{n} \times [(\vec{n} - \vec{\beta}) \times \dot{\vec{\beta}}]|^2}{(1 - \vec{\beta} \cdot \vec{n})^5}$$

and the total radiation power at t_r is then

$$P(t_r) = \frac{q^2}{4\pi c} \int \frac{|\vec{n} \times [(\vec{n} - \vec{\beta}) \times \dot{\vec{\beta}}]|^2}{(1 - \vec{\beta} \cdot \vec{n})^5} d\Omega$$

where

$$\begin{aligned} |\vec{n} \times [(\vec{n} - \vec{\beta}) \times \dot{\vec{\beta}}]|^2 &= |(\vec{n} \cdot \dot{\vec{\beta}})(\vec{n} - \vec{\beta}) - (1 - \vec{n} \cdot \vec{\beta})\dot{\vec{\beta}}|^2 \\ &= (\vec{n} \cdot \dot{\vec{\beta}})^2(1 + \beta^2 - 2\vec{n} \cdot \vec{\beta}) + (1 - \vec{n} \cdot \vec{\beta})^2\dot{\beta}^2 \\ &\quad - 2(1 - \vec{n} \cdot \vec{\beta}) \left[(\vec{n} \cdot \dot{\vec{\beta}})^2 - (\vec{n} \cdot \dot{\vec{\beta}})(\vec{\beta} \cdot \dot{\vec{\beta}}) \right] \\ &= (\vec{n} \cdot \dot{\vec{\beta}})^2(\beta^2 - 1) + (1 - \vec{n} \cdot \vec{\beta})^2\dot{\beta}^2 + 2(1 - \vec{n} \cdot \vec{\beta})(\vec{n} \cdot \dot{\vec{\beta}})(\vec{\beta} \cdot \dot{\vec{\beta}}) \end{aligned}$$

To evaluate the integral over the solid angle for $P(t_r)$, we choose a coordinate such that

$$\begin{aligned} \vec{\beta} &= \beta \vec{e}_z \\ \dot{\vec{\beta}} &= \dot{\beta}_x \vec{e}_x + \dot{\beta}_y \vec{e}_y + \dot{\beta}_z \vec{e}_z \\ \vec{n} &= \sin \theta \cos \phi \vec{e}_x + \sin \theta \sin \phi \vec{e}_y + \cos \theta \vec{e}_z \end{aligned}$$

The 1st term in the integral for $P(t_r)$ is then

$$\begin{aligned}
& \int_0^\pi \sin \theta d\theta \int_0^{2\pi} d\phi \frac{(\vec{n} \cdot \dot{\vec{\beta}})^2 (\beta^2 - 1)}{(1 - \vec{\beta} \cdot \vec{n})^5} \\
&= (\beta^2 - 1) \int_0^\pi \sin \theta d\theta \int_0^{2\pi} d\phi \frac{(\dot{\beta}_x \sin \theta \cos \phi + \dot{\beta}_y \sin \theta \sin \phi + \dot{\beta}_z \cos \theta)^2}{(1 - \beta \cos \theta)^5} \\
&= -\frac{\pi}{\gamma^2} \int_0^\pi \frac{(\dot{\beta}_x^2 + \dot{\beta}_y^2) \sin^2 \theta + 2\dot{\beta}_z^2 \cos^2 \theta}{(1 - \beta \cos \theta)^5} \sin \theta d\theta \\
&= -\frac{\pi}{\gamma^2} \left[(\dot{\beta}_x^2 + \dot{\beta}_y^2) \int_{-1}^1 \frac{dx}{(1 - \beta x)^5} - (\dot{\beta}_x^2 + \dot{\beta}_y^2 - 2\dot{\beta}_z^2) \int_{-1}^1 \frac{x^2 dx}{(1 - \beta x)^5} \right] \\
&= -\frac{\pi}{\gamma^2} \left[2\gamma^8 (1 + \beta^2) (\dot{\beta}_x^2 + \dot{\beta}_y^2) - \frac{2}{3} \gamma^8 (1 + 5\beta^2) (\dot{\beta}_x^2 + \dot{\beta}_y^2 - 2\dot{\beta}_z^2) \right] \\
&= -\frac{4\pi}{3} \gamma^6 \left[(1 - \beta^2) (\dot{\beta}_x^2 + \dot{\beta}_y^2) + (1 + 5\beta^2) \dot{\beta}_z^2 \right] \\
&= -\frac{4\pi}{3} \gamma^4 \left(\dot{\beta}^2 + 6\gamma^2 \beta^2 \dot{\beta}_z^2 \right) = -\frac{4\pi}{3} \gamma^4 \left[\dot{\beta}^2 + 6\gamma^2 (\vec{\beta} \cdot \dot{\vec{\beta}})^2 \right]
\end{aligned}$$

The 2nd term in the integral for $P(t_r)$ is

$$\int_0^\pi \sin \theta d\theta \int_0^{2\pi} d\phi \frac{(1 - \vec{n} \cdot \vec{\beta})^2 \dot{\beta}^2}{(1 - \vec{\beta} \cdot \vec{n})^5} = 2\pi \dot{\beta}^2 \int_{-1}^1 \frac{dx}{(1 - \beta x)^3} = 4\pi \gamma^4 \dot{\beta}^2$$

and the 3rd term in the integral for $P(t_r)$ is

$$\begin{aligned}
& \int_0^\pi \sin \theta d\theta \int_0^{2\pi} d\phi \frac{2(1 - \vec{n} \cdot \vec{\beta})(\vec{n} \cdot \dot{\vec{\beta}})(\vec{\beta} \cdot \dot{\vec{\beta}})}{(1 - \vec{\beta} \cdot \vec{n})^5} \\
&= 2(\vec{\beta} \cdot \dot{\vec{\beta}}) \int_0^\pi \sin \theta d\theta \int_0^{2\pi} d\phi \frac{\dot{\beta}_x \sin \theta \cos \phi + \dot{\beta}_y \sin \theta \sin \phi + \dot{\beta}_z \cos \theta}{(1 - \beta \cos \theta)^4} \\
&= 4\pi (\vec{\beta} \cdot \dot{\vec{\beta}}) \dot{\beta}_z \int_{-1}^1 \frac{x dx}{(1 - \beta x)^4} = \frac{32\pi}{3} \gamma^6 (\vec{\beta} \cdot \dot{\vec{\beta}})^2
\end{aligned}$$

The total radiation power is then

$$\begin{aligned}
P(t_r) &= \frac{q^2}{4\pi c} \left\{ -\frac{4\pi}{3} \gamma^4 \left[\dot{\beta}^2 + 6\gamma^2 (\vec{\beta} \cdot \dot{\vec{\beta}})^2 \right] + 4\pi \gamma^4 \dot{\beta}^2 + \frac{32\pi}{3} \gamma^6 (\vec{\beta} \cdot \dot{\vec{\beta}})^2 \right\} \\
&= \frac{2q^2}{3c} \gamma^4 \left[\dot{\beta}^2 + \gamma^2 (\vec{\beta} \cdot \dot{\vec{\beta}})^2 \right]
\end{aligned}$$

Note that this is the general formula for the total radiation power. For $\beta \ll 1$, it is reduced to Larmor formula,

$$P = \frac{2q^2}{3c^3} \dot{v}^2$$

To write $P(t_r)$ in the covariant form, considering $(p_0, \vec{p}) = \gamma(mc, m\vec{v})$ and $dt = \gamma d\tau$,

$$\frac{d\vec{p}}{dt} = \gamma mc \frac{d\vec{\beta}}{dt} + mc\vec{\beta} \frac{d\gamma}{dt}$$

Thus

$$\frac{d\vec{\beta}}{dt} = \frac{1}{\gamma mc} \left(\frac{d\vec{p}}{d\tau} - mc\vec{\beta} \frac{d\gamma}{d\tau} \right) \frac{d\tau}{dt} = \frac{1}{\gamma^2 mc} \left(\frac{d\vec{p}}{d\tau} - mc\vec{\beta} \frac{d\gamma}{d\tau} \right)$$

and

$$\begin{aligned} \left[\dot{\beta}^2 + \gamma^2 (\vec{\beta} \cdot \dot{\vec{\beta}})^2 \right] &= \frac{1}{\gamma^4 m^2 c^2} \left[\frac{d\vec{p}}{d\tau} \cdot \frac{d\vec{p}}{d\tau} + m^2 c^2 \beta^2 \left(\frac{d\gamma}{d\tau} \right)^2 - 2mc \left(\vec{\beta} \cdot \frac{d\vec{p}}{d\tau} \right) \frac{d\gamma}{d\tau} \right] \\ &\quad + \frac{1}{\gamma^2 m^2 c^2} \left(\vec{\beta} \cdot \frac{d\vec{p}}{d\tau} - mc\beta^2 \frac{d\gamma}{d\tau} \right)^2 \\ &= \frac{1}{\gamma^4 m^2 c^2} \left[\frac{d\vec{p}}{d\tau} \cdot \frac{d\vec{p}}{d\tau} + m^2 c^2 \beta^2 \left(\frac{d\gamma}{d\tau} \right)^2 - 2mc \left(\vec{\beta} \cdot \frac{d\vec{p}}{d\tau} \right) \frac{d\gamma}{d\tau} \right. \\ &\quad \left. + \gamma^2 \left(\vec{\beta} \cdot \frac{d\vec{p}}{d\tau} \right)^2 + m^2 c^2 \beta^4 \gamma^2 \left(\frac{d\gamma}{d\tau} \right)^2 - 2mc\beta^2 \gamma^2 \left(\vec{\beta} \cdot \frac{d\vec{p}}{d\tau} \right) \frac{d\gamma}{d\tau} \right] \\ &= \frac{1}{\gamma^4 m^2 c^2} \left[\frac{d\vec{p}}{d\tau} \cdot \frac{d\vec{p}}{d\tau} + \beta^2 \gamma^2 \left(\frac{d\gamma mc}{d\tau} \right)^2 - 2\gamma^2 \left(\vec{\beta} \cdot \frac{d\vec{p}}{d\tau} \right) \frac{d\gamma mc}{d\tau} + \left(\gamma \vec{\beta} \cdot \frac{d\vec{p}}{d\tau} \right)^2 \right] \\ &= \frac{1}{\gamma^4 m^2 c^2} \left[\frac{d\vec{p}}{d\tau} \cdot \frac{d\vec{p}}{d\tau} - \frac{dp_0}{d\tau} \cdot \frac{dp_0}{d\tau} + \gamma^2 \left(\frac{dp_0}{d\tau} \right)^2 - 2\gamma^2 \left(\vec{\beta} \cdot \frac{d\vec{p}}{d\tau} \right) \frac{dp_0}{d\tau} + \left(\gamma \vec{\beta} \cdot \frac{d\vec{p}}{d\tau} \right)^2 \right] \\ &= \frac{1}{\gamma^4 m^2 c^2} \left[-\frac{dp_\alpha}{d\tau} \frac{dp^\alpha}{d\tau} + \left(\gamma \frac{dp_0}{d\tau} - \gamma \vec{\beta} \cdot \frac{d\vec{p}}{d\tau} \right)^2 \right] \\ &= \frac{1}{\gamma^4 m^2 c^2} \left[\frac{1}{m^2 c^2} \left(p_\alpha \frac{dp_\alpha}{d\tau} \right)^2 - \frac{dp_\alpha}{d\tau} \frac{dp^\alpha}{d\tau} \right] \end{aligned}$$

The total radiation power can be written in the covariant form as

$$P(t_r) = \frac{2q^2}{3m^2 c^3} \left[\left(p_\alpha \frac{dp_\alpha}{d\tau} \right)^2 - \frac{dp_\alpha}{d\tau} \frac{dp^\alpha}{d\tau} \right] = \frac{2q^2}{3m^2 c^3} \left[\frac{dp_\alpha}{d\tau} (p^\alpha p_\beta - \delta^\alpha_\beta) \frac{dp^\beta}{d\tau} \right]$$

For a charge moving in external electric and magnetic field, the equation of motion is

$$\frac{d\vec{p}}{dt} = q \left(\vec{E} + \vec{\beta} \times \vec{B} \right) \quad \text{and} \quad \frac{dp_0}{dt} = q\vec{\beta} \cdot \vec{E}$$

Thus

$$p_\alpha \frac{dp_\alpha}{d\tau} = \gamma \left(p_0 \frac{dp_0}{dt} - \vec{p} \cdot \frac{d\vec{p}}{dt} \right) = q\gamma \left[\gamma mc (\vec{\beta} \cdot \vec{E}) - \gamma m\vec{v} \cdot (\vec{E} + \vec{\beta} \times \vec{B}) \right] = 0$$

and

$$-\frac{dp_\alpha}{d\tau} \frac{dp^\alpha}{d\tau} = q^2 \gamma^2 \left[(\vec{E} + \vec{\beta} \times \vec{B})^2 - (\vec{\beta} \cdot \vec{E})^2 \right]$$

Therefore,

$$P(t_r) = \frac{2q^4}{3m^2 c^3} \gamma^2 \left[(\vec{E} + \vec{\beta} \times \vec{B})^2 - (\vec{\beta} \cdot \vec{E})^2 \right]$$

(b) The covariant form of equation of motion in a magnetic and electric field can be written as

$$\frac{dp^\alpha}{d\tau} = \frac{q}{c} F^{\alpha\beta} U_\beta = \frac{q}{mc} F^{\alpha\beta} p_\beta$$

Therefore, the radiation power of a charged particle in an external magnetic and electric field can be expressed as

$$P(t_r) = -\frac{2q^2}{3m^2 c^3} \left(\frac{dp_\alpha}{d\tau} \frac{dp^\alpha}{d\tau} \right) = -\frac{2q^4}{3m^4 c^5} F_{\alpha\nu} p^\nu F^{\alpha\beta} p_\beta = \frac{2q^4}{3m^4 c^5} F^{\alpha\beta} p_\beta p^\nu F_{\nu\alpha}$$

where $F_{\nu\alpha} = -F_{\alpha\nu}$.

14.12 Similar to the Problem 14.4 (but not necessarily $\beta \ll 1$), a particle moving along the z axis with

$$z(t) = a \cos \omega_0 t$$

(a) The radiation power per solid angle is

$$\frac{dP(t')}{d\Omega} = \frac{e^2}{4\pi c} \frac{|\vec{n} \times [(\vec{n} - \vec{\beta}) \times \dot{\vec{\beta}}]|^2}{(1 - \vec{\beta} \cdot \vec{n})^5}$$

where

$$\vec{\beta} = \frac{dz}{cdt} \vec{e}_z = -\left(\frac{a\omega_0}{c} \sin \omega_0 t\right) \vec{e}_z \quad \text{and} \quad \dot{\vec{\beta}} = \frac{d\vec{\beta}}{dt} = -\left(\frac{a\omega_0^2}{c} \cos \omega_0 t\right) \vec{e}_z$$

Let

$$\begin{aligned} \vec{n} &= (\cos \phi \sin \theta) \vec{e}_x + (\sin \phi \sin \theta) \vec{e}_y + (\cos \theta) \vec{e}_z \\ \vec{n} \times [(\vec{n} - \vec{\beta}) \times \dot{\vec{\beta}}] &= \vec{n} \times (\vec{n} \times \dot{\vec{\beta}}) = (\vec{n} \cdot \dot{\vec{\beta}}) \vec{n} - \dot{\vec{\beta}} \end{aligned}$$

and

$$\begin{aligned} |\vec{n} \times [(\vec{n} - \vec{\beta}) \times \dot{\vec{\beta}}]|^2 &= \left| \vec{n} \times (\vec{n} \times \dot{\vec{\beta}}) \right|^2 \\ &= (\vec{n} \cdot \dot{\vec{\beta}})^2 + \dot{\vec{\beta}}^2 - 2(\vec{n} \cdot \dot{\vec{\beta}}) \vec{n} \cdot \dot{\vec{\beta}} \\ &= \dot{\vec{\beta}}^2 - (\vec{n} \cdot \dot{\vec{\beta}})^2 = \dot{\vec{\beta}}^2 \sin^2 \theta \end{aligned}$$

Thus

$$\frac{dP(t')}{d\Omega} = \frac{e^2 a^2 \omega_0^4}{4\pi c^3} \frac{\sin^2 \theta \cos^2 \omega_0 t'}{[1 + (a\omega_0/c) \cos \theta \sin \omega_0 t']^5}$$

(b) The average power over a period of particle oscillation $T = 2\pi/\omega_0$ is

$$\begin{aligned} \left\langle \frac{dP(t')}{d\Omega} \right\rangle_{t'} &= \frac{1}{T} \int_0^T \frac{dP(t')}{d\Omega} dt = \frac{e^2 a^2 \omega_0^5}{2(2\pi)^2 c^3} \sin^2 \theta \int_0^{2\pi/\omega_0} \frac{\cos^2 \omega_0 t}{[1 + (a\omega_0/c) \cos \theta \sin \omega_0 t]^5} dt \\ &= \frac{e^2 a^2 \omega_0^4}{(2\pi)^2 c^3} \sin^2 \theta \int_{-1}^1 \frac{\sqrt{1 - \sin^2 \psi}}{(1 + b \sin \psi)^5} d \sin \psi = \frac{e^2 a^2 \omega_0^4}{(2\pi)^2 c^3} \sin^2 \theta \int_{-1}^1 \frac{\sqrt{1 - x^2}}{(1 + bx)^5} dx \end{aligned}$$

where $b = (a\omega_0/c) \cos \theta$.

HW Assignment 14-4 (14.14, 14.26)

14.14 Continue Problem 14.2.

For the charged particle moving in the periodic oscillation in Problem 14.2, the instantaneous radiation power is

$$\frac{dP(t)}{d\Omega} = \left| \sqrt{\frac{c}{4\pi}} R \vec{E} \right|^2 = \left| \sqrt{\frac{c}{4\pi}} R \vec{B} \right|^2$$

where $\vec{B}(t) = \vec{n} \times \vec{E}$. Let

$$\vec{A}(t) = \sqrt{\frac{c}{4\pi}} R \vec{E} \quad \text{and} \quad \vec{C}(t) = \sqrt{\frac{c}{4\pi}} R \vec{B}$$

In the case that the charged particle moves in a periodic motion with a fundamental frequency ω_0 , the Fourier transformation of $\vec{A}(t)$ has a discrete spectrum containing frequencies of $m\omega_0$, where m is any integer.

$$\vec{A}_m = \frac{1}{T} \int_0^T \vec{A}(t) e^{im\omega_0 t} dt, \quad \vec{A}(t) = \sum_{m=-\infty}^{\infty} \vec{A}_m e^{-im\omega_0 t}$$

and

$$\vec{C}_m = \frac{1}{T} \int_0^T \vec{C}(t) e^{im\omega_0 t} dt, \quad \vec{C}(t) = \sum_{m=-\infty}^{\infty} \vec{C}_m e^{-im\omega_0 t}$$

where $T = 2\pi/\omega_0$.

$$\begin{aligned} \left\langle \frac{dP(t')}{d\Omega} \right\rangle_t &= \frac{1}{T} \int_0^T \frac{dP(t')}{d\Omega} dt = \frac{1}{T} \int_0^T |\vec{A}(t)|^2 dt = \frac{1}{T} \int_0^T dt \left[\sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} \vec{A}_n^* \vec{A}_m e^{i(n-m)\omega_0 t} \right] \\ &= \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} \vec{A}_n^* \vec{A}_m \left[\frac{1}{T} \int_0^T e^{i(n-m)\omega_0 t} dt \right] \\ &= \sum_{m=-\infty}^{\infty} \sum_{n=-\infty}^{\infty} \vec{A}_n^* \vec{A}_m \delta_{mn} = \sum_{m=-\infty}^{\infty} |\vec{A}_m|^2 = 2 \sum_{m=0}^{\infty} |\vec{A}_m|^2 \end{aligned}$$

$$\begin{aligned}
\vec{A}_m &= \frac{1}{T} \int_0^T \vec{A}(t) e^{im\omega_0 t} dt = \frac{1}{T} \left(\frac{e^2}{4\pi c} \right)^{1/2} \int_0^T \left\{ \frac{\vec{n} \times [(\vec{n} - \vec{\beta}) \times \dot{\vec{\beta}}]}{(1 - \vec{\beta} \cdot \vec{n})^3} \right\}_{ret} e^{im\omega_0 t} dt \\
&= \frac{1}{T} \left(\frac{e^2}{4\pi c} \right)^{1/2} e^{im\omega_0(\vec{x} \cdot \vec{n})/c} \int_0^T \frac{\vec{n} \times [(\vec{n} - \vec{\beta}) \times \dot{\vec{\beta}}]}{(1 - \vec{\beta} \cdot \vec{n})^2} e^{im\omega_0[t' - \vec{n} \cdot \vec{r}(t')/c]} dt' \\
&= i \frac{m\omega_0}{T} \left(\frac{e^2}{4\pi c} \right)^{1/2} e^{im\omega_0(\vec{x} \cdot \vec{n})/c} \int_0^T \left\{ \vec{n} \times [\vec{n} \times \vec{\beta}(t')] \right\} e^{im\omega_0[t - \vec{n} \cdot \vec{r}(t')/c]} dt' \\
&= i \frac{m\omega_0^2}{2\pi} \left(\frac{e^2}{4\pi c} \right)^{1/2} e^{im\omega_0(\vec{x} \cdot \vec{n})/c} \int_0^T \left\{ \vec{n} \times [\vec{n} \times \vec{\beta}(t')] \right\} e^{im\omega_0[t - \vec{n} \cdot \vec{r}(t')/c]} dt'
\end{aligned}$$

and

$$\vec{C}_m = i \frac{m\omega_0^2}{2\pi} \left(\frac{e^2}{4\pi c} \right)^{1/2} e^{im\omega_0(\vec{x} \cdot \vec{n})/c} \int_0^T [\vec{n} \times \vec{\beta}(t')] e^{im\omega_0[t - \vec{n} \cdot \vec{r}(t')/c]} dt'$$

The average power radiated per unit solid angle in the m th harmonic is then

$$\frac{dP_m}{d\Omega} = 2|A_m|^2 = \frac{m^2\omega_0^4 e^2}{(2\pi c)^3} \left| \int_0^T \{ \vec{n} \times (\vec{n} \times \vec{v}) \} e^{im\omega_0(t - \vec{n} \cdot \vec{r}/c)} dt \right|^2$$

or

$$\frac{dP_m}{d\Omega} = 2|C_m|^2 = \frac{m^2\omega_0^4 e^2}{(2\pi c)^3} \left| \int_0^T (\vec{n} \times \vec{v}) e^{im\omega_0(t - \vec{n} \cdot \vec{r}/c)} dt \right|^2$$

This is the result of Problem 14.13. For the particle oscillating in the z direction,

$$\vec{r}(t) = (a \cos \omega_0 t) \vec{e}_z \quad \text{and} \quad \vec{v} = -(a\omega_0 \sin \omega_0 t) \vec{e}_z$$

Let

$$\vec{n} = (\cos \phi \sin \theta) \vec{e}_x + (\sin \phi \sin \theta) \vec{e}_y + (\cos \theta) \vec{e}_z$$

$$\vec{n} \times \vec{v} = v[(\sin \phi \sin \theta) \vec{e}_x - (\cos \phi \sin \theta) \vec{e}_y] \quad \text{and} \quad \vec{n} \cdot \vec{r} = a \cos \omega_0 t \cos \theta$$

Then,

$$\begin{aligned}
\frac{dP_m}{d\Omega} &= \frac{m^2\omega_0^4 e^2}{(2\pi c)^3} \left| \int_0^T (\vec{n} \times \vec{v}) e^{im\omega_0(t - \vec{n} \cdot \vec{r}/c)} dt \right|^2 \\
&= \frac{m^2\omega_0^4 e^2}{(2\pi c)^3} |(\sin \phi \sin \theta) \vec{e}_x - (\cos \phi \sin \theta) \vec{e}_y|^2 \left| \int_0^T v(t) e^{im\omega_0(t - \vec{n} \cdot \vec{r}/c)} dt \right|^2 \\
&= \frac{m^2 a^2 \omega_0^4 e^2}{(2\pi c)^3} \sin^2 \theta \left| \int_0^{2\pi} \sin \psi \exp \left[im \left(\psi - \frac{a\omega_0 \cos \theta}{c} \cos \psi \right) \right] d\psi \right|^2
\end{aligned}$$

(b) For nonrelativistic case,

$$\frac{a\omega_0}{c} \ll 1$$

and

$$\begin{aligned} \frac{dP_m}{d\Omega} &= \frac{m^2 a^2 \omega_0^4 e^2}{(2\pi c)^3} \sin^2 \theta \left| \int_0^{2\pi} \sin \psi e^{im\psi} d\psi \right|^2 = \frac{m^2 a^2 \omega_0^4 e^2}{8\pi c^3} \sin^2 \theta \delta_{m1} \\ \frac{dP}{d\Omega} &= \sum_{m=0}^{\infty} \frac{dP_m}{d\Omega} = \frac{a^2 \omega_0^4 e^2}{8\pi c^3} \sin^2 \theta \end{aligned}$$

and

$$P = \frac{a^2 \omega_0^4 e^2}{8\pi c^3} \int_0^{\pi} \int_0^{2\pi} \sin^2 \theta \sin \theta d\theta d\phi = \frac{a^2 \omega_0^4 e^2}{3c^3}$$

14.26 Synchrotron radiation of electrons in Crab nebula.

(a) $E = 10^{13}$ eV and $B = 3 \times 10^{-4}$ gauss = 3×10^{-8} T.

For the circular motion, since

$$\gamma m \frac{v^2}{\rho} = \frac{e}{c} v B,$$

the radius of the electron circular orbit is

$$\rho = \frac{c\gamma m v}{eB} = \frac{cp}{eB} = \frac{\sqrt{E^2 - (mc^2)^2}}{eB} \simeq \frac{E}{eB} = \frac{10^{13} \text{ eV}}{e \times 3 \times 10^{-8}} = 3 \times 10^{20} \text{ m}$$

where $mc^2 = 0.511 \times 10^6$ eV for electron and $E \gg mc^2$. The fundamental frequency is the frequency of the circular motion for the electron which is

$$\omega_0 = \omega_B = \frac{v}{\rho} = \frac{eB}{\gamma mc} \simeq 10^{-12} \text{ Hz}$$

where $\gamma = E/mc^2 = 2 \times 10^7$. The critical frequency is

$$\omega_c = 3\gamma^3 \omega_B = 3\gamma^2 \frac{eB}{mc} \simeq 2 \times 10^{10} \text{ Hz}$$

(b)

The total radiation energy per unit frequency for a charged particle in a circular motion is

$$\frac{dE}{d\omega} = \int \frac{d^2 E}{d\omega d\Omega} d\Omega \sim \left. \frac{d^2 E}{d\omega d\Omega} \right|_{\theta=0} (2\pi\theta_c)$$

where, from the Lecture Note, we have

$$\left. \frac{d^2 E}{d\omega d\Omega} \right|_{\theta=0} \simeq \begin{cases} \left(\frac{3e^2}{4\pi c} \right) \gamma^2 \frac{\omega}{\omega_c} e^{-\omega/\omega_c} & \text{for } \omega \gg \omega_c \\ \frac{3e^2}{4^{1/3} c} \left[\frac{\Gamma(2/3)}{\pi} \right]^2 \gamma^2 \left(\frac{\omega}{\omega_c} \right)^{2/3} & \text{for } \omega \ll \omega_c \end{cases}$$

and

$$\theta_c \simeq \begin{cases} \frac{1}{\gamma} \left(\frac{3\omega}{\omega_c} \right)^{-1/2} & \text{for } \omega \gg \omega_c \\ \frac{1}{\gamma} \left(\frac{2\omega}{\omega_c} \right)^{-1/3} & \text{for } \omega \ll \omega_c \end{cases}$$

For a periodic circular motion, the radiation is discrete with harmonic frequency of $n\omega_B$, where n is an integer. For the discrete Fourier transformation, $1/(2\pi)$ in the front of the transformation integral should be changed into $1/T$, where $T = 2\pi/\omega_B$ is the period of the periodic motion. Moreover, since the total radiation energy per unit frequency during each circle of the motion is

$$\frac{dE}{d\omega} = \int_0^T \frac{dP}{d\omega} dt \sim T \frac{dP}{d\omega} \quad \Rightarrow \quad \frac{dP}{d\omega} \sim \frac{1}{T} \frac{dE}{d\omega}$$

Therefore, the radiation power per unit frequency from a charged particle in a circular motion is

$$\frac{dP}{d\omega} \simeq \frac{1}{T^2} \frac{dE}{d\omega} \simeq \frac{\omega_B^2}{2\pi} \frac{d^2 E}{d\omega d\Omega} \Big|_{\theta=0} \theta_c \simeq \begin{cases} \left(\frac{\sqrt{3}e^2}{8\pi^2 c} \right) \omega_B^2 \gamma \left(\frac{\omega}{\omega_c} \right)^{1/2} e^{-\omega/\omega_c} & \text{for } \omega \gg \omega_c \\ \frac{3e^2}{4\pi c} \left[\frac{\Gamma(2/3)}{\pi} \right]^2 \omega_B^2 \gamma \left(\frac{\omega}{\omega_c} \right)^{1/3} & \text{for } \omega \ll \omega_c \end{cases}$$

Considering the γ -dependence, $\omega_B \sim 1/\gamma$ and $\omega_c \sim \gamma^3 \omega_B \sim \gamma^2$. Therefore,

$$\frac{dP}{d\omega} \sim \begin{cases} \left(\frac{\omega}{\gamma^4} \right)^{1/2} e^{-\omega/\omega_c} \sim \left(\frac{\omega}{E^4} \right)^{1/2} e^{-\omega/\omega_c} & \text{for } \omega \gg \omega_c \\ \left(\frac{\omega}{\gamma^5} \right)^{1/3} \sim \left(\frac{\omega}{E^5} \right)^{1/3} & \text{for } \omega \ll \omega_c \end{cases}$$

where $E = \gamma mc^2$ is the energy of the charged particle.

(c) Since each photon carries an energy of $\hbar\omega$, the number of photon in the unit frequency interval is

$$\mathcal{N}(\omega) d\omega = \frac{1}{\hbar\omega} \left(\frac{dE}{d\omega} \right) d\omega$$

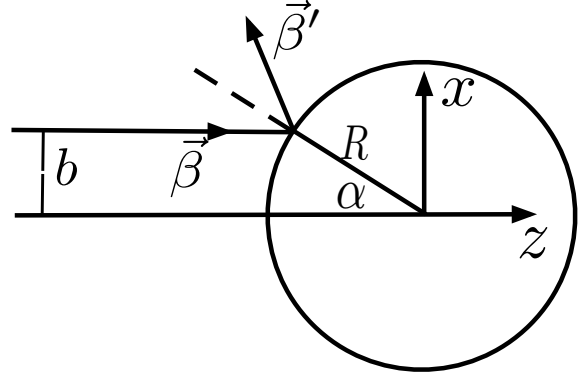
HW Assignment 7 (15.2, 15.4)

15.2 A nonrelativistic particle of charge e and mass m collides elastically with a fixed, smooth, hard sphere of radius R .

For a uniform beam of the incident particles with different impact parameter, the classical differential cross section for the emission of photons per solid angle per unit energy interval is defined as

$$\frac{d^2\sigma}{d\Omega d(\hbar\omega)} = \int b db d\psi \frac{1}{\hbar^2\omega} \left(\frac{d^2W}{d\Omega d\omega} \right)$$

Note that $b db d\psi$ is the area element on the cross-section of the incident beam. As shown in the figure, the impact parameter is related to the incident angle as



$$b = R \sin \alpha \quad \text{and} \quad db = R \cos \alpha d\alpha$$

Then

$$\frac{d^2\sigma}{d\Omega d(\hbar\omega)} = \frac{R^2}{\hbar^2\omega} \int_0^{\pi/2} \sin \alpha \cos \alpha d\alpha \int_0^{2\pi} d\psi \left(\frac{d^2W}{d\Omega d\omega} \right)$$

For the hard (head-on) collision (see Lecture Note page 15.1), the radiation power spectrum is

$$\frac{d^2W}{d\Omega d\omega} = \frac{e^2}{4\pi^2c} \left| \left[\frac{\vec{n} \times (\vec{n} \times \vec{\beta})}{1 - \vec{\beta} \cdot \vec{n}} \right]_{after} - \left[\frac{\vec{n} \times (\vec{n} \times \vec{\beta})}{1 - \vec{\beta} \cdot \vec{n}} \right]_{before} \right|^2$$

With $\beta \ll 1$,

$$\frac{d^2W}{d\Omega d\omega} = \frac{e^2}{4\pi^2c} \left| \vec{n} \times [\vec{n} \times (\vec{\beta} - \vec{\beta}')] \right|^2 = \frac{e^2}{4\pi^2c} \left\{ |\vec{\beta} - \vec{\beta}'|^2 - [\vec{n} \cdot (\vec{\beta} - \vec{\beta}')]^2 \right\}$$

where $\vec{\beta}'$ is the velocity of the particle after the collision. If we choose $\vec{e}_x$ along the horizontal and $\vec{e}_y$ along the vertical as shown in the figure and let

$$\vec{n} = \cos \phi \sin \theta \vec{e}_x + \sin \phi \sin \theta \vec{e}_y + \cos \theta \vec{e}_z,$$

then

$$\begin{aligned} \vec{\beta} &= \beta \vec{e}_z \\ \vec{\beta}' &= \beta \sin 2\alpha \vec{e}_x - \beta \cos 2\alpha \vec{e}_z \\ \vec{\beta} - \vec{\beta}' &= -\beta \sin 2\alpha \vec{e}_x + \beta(1 + \cos 2\alpha) \vec{e}_z \\ |\vec{\beta} - \vec{\beta}'|^2 &= \beta^2 (1 + 2 \cos 2\alpha) \\ [\vec{n} \cdot (\vec{\beta} - \vec{\beta}')]^2 &= \beta^2 [\cos \phi \sin \theta \sin 2\alpha - \cos \theta (1 + \cos 2\alpha)]^2 \end{aligned}$$

Since

$$\int_0^{\pi/2} \sin \alpha \cos \alpha d\alpha \int_0^{2\pi} d\psi \left| \vec{\beta} - \vec{\beta}' \right|^2 = \pi \beta^2 \int_0^{\pi/2} (1 + 2 \cos 2\alpha) \sin 2\alpha d\alpha = \pi \beta^2$$

$$\int_0^{\pi/2} \sin \alpha \cos \alpha d\alpha \int_0^{2\pi} d\psi \left[\vec{n} \cdot (\vec{\beta} - \vec{\beta}') \right]^2 = -\pi \beta^2 \int_0^{\pi} [\cos \phi \sin \theta \sin \xi - \cos \theta (1 + \cos \xi)]^2 d \cos \xi$$

15.4 A group of charged particle with charges e_j and $\vec{r}_j(t)$ are accelerated during $-\tau/2 < t < \tau/2$ and their velocities change from $-c\vec{\beta}_j$ to $c\vec{\beta}_j$. Obtain $d^2E/d\omega d\Omega$.

From the lecture note,

$$\frac{d^2E}{d\Omega d\omega} = \frac{1}{\pi} \left| \vec{A}(\omega) \right|^2$$

where

$$\vec{A}(\omega) = \frac{q}{\sqrt{4\pi c}} e^{i\omega(\vec{x} \cdot \vec{n})/c} \int_{-\tau/2}^{\tau/2} \frac{\vec{n} \times [(\vec{n} - \vec{\beta}) \times \dot{\vec{\beta}}]}{(1 - \vec{n} \cdot \vec{\beta})^2} e^{i\omega[t - \vec{n} \cdot \vec{r}(t)/c]} dt$$

$$\stackrel{\dot{\vec{n}} \simeq 0}{=} \frac{q}{\sqrt{4\pi c}} e^{i\omega(\vec{x} \cdot \vec{n})/c} \int_{-\tau/2}^{\tau/2} \frac{d}{dt} \left[\frac{\vec{n} \times (\vec{n} \times \vec{\beta})}{(1 - \vec{n} \cdot \vec{\beta})} \right] e^{i\omega[t - \vec{n} \cdot \vec{r}(t)/c]} dt$$

where

$$\vec{n} \times (\vec{n} \times \vec{\beta}) = \vec{n}(\vec{n} \cdot \vec{\beta}) - \vec{\beta} = -\vec{\beta}_{\perp}$$

(a) For $\omega\tau \ll 1$,

$$\vec{A}(\omega) \stackrel{\dot{\vec{n}} \simeq 0}{=} \frac{q}{\sqrt{4\pi c}} e^{i\omega(\vec{x} \cdot \vec{n})/c} \int_{-\tau/2}^{\tau/2} \frac{d}{dt} \left[\frac{\vec{n} \times (\vec{n} \times \vec{\beta})}{(1 - \vec{n} \cdot \vec{\beta})} \right] e^{i\omega[t - \vec{n} \cdot \vec{r}(t)/c]} dt$$

$$\stackrel{\omega\tau \ll 1}{=} \frac{q}{\sqrt{4\pi c}} e^{i\omega(\vec{x} \cdot \vec{n})/c} e^{-i\omega\vec{n} \cdot \vec{r}(0)/c} \int_{-\tau/2}^{\tau/2} \frac{d}{dt} \left[\frac{\vec{n} \times (\vec{n} \times \vec{\beta})}{(1 - \vec{n} \cdot \vec{\beta})} \right] dt$$

$$= -\frac{q}{\sqrt{4\pi c}} e^{i\omega(\vec{x} \cdot \vec{n})/c} e^{-i\omega\vec{n} \cdot \vec{r}(0)/c} \left(\frac{\vec{\beta}'_{\perp}}{1 - \vec{n} \cdot \vec{\beta}'} - \frac{\vec{\beta}_{\perp}}{1 - \vec{n} \cdot \vec{\beta}} \right)$$

For a group of particles, then

$$\frac{d^2E}{d\Omega d\omega} = \frac{1}{4\pi^2 c} \left| \sum_j q_j e^{-i\omega\vec{n} \cdot \vec{r}_j(0)/c} \left(\frac{\vec{\beta}'_{j\perp}}{1 - \vec{n} \cdot \vec{\beta}'_j} - \frac{\vec{\beta}_{j\perp}}{1 - \vec{n} \cdot \vec{\beta}_j} \right) \right|^2$$

(b) For the decay of an ω^0 meson into a pair of $\pi^+ \pi^-$ or $e^+ e^-$, two particles with opposite charges are created at $t = 0$ and at $\vec{r}(0) = 0$, and their velocities are increased from 0 at $t = 0$ to $c\beta$ at $t = \tau$, where $\tau \sim 0$ ($\tau E \sim h$). In the center-of-mass frame of the particles (the frame moving with ω^0 meson), the two opposite charges move toward opposite directions, i.e. $\vec{\beta}_1 = -\vec{\beta}_2$,

$$-\vec{\beta}_2 \cdot \vec{n} = \vec{\beta}_1 \cdot \vec{n} = \beta \cos \theta$$

and for $\omega\tau \ll 1$

$$\begin{aligned} \sum_j q_j e^{-i\omega\vec{n}\cdot\vec{r}_j(0)/c} \left(\frac{\vec{\beta}'_{j\perp}}{1 - \vec{n} \cdot \vec{\beta}'_j} - \frac{\vec{\beta}_{j\perp}}{1 - \vec{n} \cdot \vec{\beta}_j} \right) &= \frac{e\vec{\beta}_{1\perp}}{1 - \vec{n} \cdot \vec{\beta}_1} - \frac{e\vec{\beta}_{2\perp}}{1 - \vec{n} \cdot \vec{\beta}_2} \\ &= e\beta \sin \theta \left(\frac{1}{1 - \beta \cos \theta} + \frac{1}{1 + \beta \cos \theta} \right) = \frac{2e\beta \sin \theta}{1 - \beta^2 \cos^2 \theta} \end{aligned}$$

Thus

$$\frac{d^2 E}{d\Omega d\omega} = \frac{e^2 \beta^2}{\pi^2 c} \frac{\sin^2 \theta}{(1 - \beta^2 \cos^2 \theta)^2}$$

and

$$\frac{dE}{d\omega} = \frac{e^2 \beta^2}{\pi^2 c} \int \frac{\sin^2 \theta}{(1 - \beta^2 \cos^2 \theta)^2} \sin \theta d\theta d\phi = \frac{2e^2}{\pi c \beta} \int_{-\beta}^{\beta} \frac{\beta^2 - x^2}{(1 - x^2)^2} dx$$

where

$$\begin{aligned} \frac{\beta^2 - x^2}{(1 - x^2)^2} &= \frac{\beta^2 - 1}{(1 - x^2)^2} + \frac{1}{1 - x^2} = \frac{\beta^2 - 1}{4} \left[\frac{1}{1 - x} + \frac{1}{1 + x} \right]^2 + \frac{1}{2(1 - x)} + \frac{1}{2(1 + x)} \\ &= \frac{\beta^2 - 1}{4} \left[\frac{1}{(1 - x)^2} + \frac{1}{(1 + x)^2} + \frac{1}{(1 - x)} + \frac{1}{(1 + x)} \right] + \frac{1}{2(1 - x)} + \frac{1}{2(1 + x)} \\ &= \frac{\beta^2 - 1}{4} \left[\frac{1}{(1 - x)^2} + \frac{1}{(1 + x)^2} \right] + \frac{\beta^2 + 1}{4} \left[\frac{1}{(1 - x)} + \frac{1}{(1 + x)} \right] \end{aligned}$$

$$\begin{aligned} \int_{-\beta}^{\beta} \frac{\beta^2 - x^2}{(1 - x^2)^2} dx &= \frac{\beta^2 - 1}{4} \left[\frac{1}{1 - x} - \frac{1}{1 + x} \right]_{-\beta}^{\beta} + \frac{\beta^2 + 1}{4} \ln \left(\frac{1 + x}{1 - x} \right) \Big|_{-\beta}^{\beta} \\ &= \frac{\beta^2 + 1}{2} \ln \left(\frac{1 + \beta}{1 - \beta} \right) - \frac{\beta}{2} \end{aligned}$$

$$\frac{dE}{d\omega} = \frac{e^2}{\pi c} \left[\frac{\beta^2 + 1}{\beta} \ln \left(\frac{1 + \beta}{1 - \beta} \right) - 1 \right] = \frac{e^2}{\pi c} \left[\frac{2\gamma^2 - 1}{\gamma\beta} \ln \left(\frac{1 + \beta}{1 - \beta} \right) - 1 \right]$$

HW Assignment for Chapter 17

13.9 (Jackson Book) Assuming that Plexiglas or Lucite has an index of refraction of $n = 1.5$ in visible region, compute the angle of emission of visible Cherenkov radiation for electrons and protons as a function of their kinetic energies in MeV. Determine how many quanta with wavelengths between 4000 to 6000 Å are emitted per centimeter of path in Lucite by a 1 MeV electron, a 500 MeV proton, and a 5 GeV proton.

Answer. The Cherenkov radiation angle is

$$\theta_C = \cos^{-1} \left(\frac{v_\phi}{v} \right) = \cos^{-1} \left(\frac{1}{n\beta} \right) = \cos^{-1} \left(\frac{\gamma}{n\sqrt{\gamma^2 - 1}} \right)$$

where $v_\beta = c/n$ and $\beta = \sqrt{\gamma^2 - 1}/\gamma$.

Extra 17.1 For the transition radiation, show that for $\beta \ll 1$,

$$\frac{dE}{d\omega} \sim (\vec{N} \cdot \vec{\beta})^2$$

and for $\beta \sim 1$,

$$\frac{dE}{d\omega} \sim \ln \gamma$$

Extra 17.2 Consider N uniformly distributed relativistic electrons moving around a circular orbit with radius R and circular frequency ω_0 .

(a) Calculate the frequency spectrum of the radiation from individual electron.

(b) Calculate the form factor of N electrons.

Extra 17.3 Consider a bunch of relativistic electrons of length L is incident to and absorbed by a conducting surface, Show that the coherent transition radiation is

$$\frac{dE}{d\Omega} \sim \frac{1}{L}$$

Answer. For a single relativistic ($\beta \sim 1$) charged particle colliding on a conductor surface, the radiation energy observed from outside the conductor is

$$\frac{d^2E}{d\omega d\Omega} = \frac{q^2}{4\pi^2 c} \left| \frac{\vec{n} \times \vec{\beta}'}{1 - \vec{n} \cdot \vec{\beta}'} \right|^2 = \frac{q^2}{4\pi^2 c} \left| \frac{\beta \sin \theta}{1 - \beta \cos \theta} \right|^2$$

where

$$1 - \vec{n} \cdot \vec{\beta}' = 1 - \beta \cos \theta \quad \text{and} \quad \vec{n} \times \vec{\beta}' = \beta \sin \theta$$